

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead



UNIT II · COMMUNICATION & NETWORKING

Topic 1

Introduction to the unit

INTERNET OF THINGS

Unit II

Communication

Why it is the backbone of IoT

Short-range | Long-range | Protocols | Networking



Communication Is the Backbone of IoT

Sensing creates the data. Communication is what carries it from the device to the cloud, and back down to the things that act.

Sense it. Move it. Use it.

In Simple Words

A sensor on its own is a silent island. Communication links each layer of the system so data can flow from where it is sensed to where it is used.

Device
Senses and sends a reading

Gateway
Gathers and forwards it onward

Cloud
Stores and analyzes it

What makes it hard?

Moving data is one of the hardest problems in all of IoT, because:

Unreliable wireless links
Data must arrive over links that can drop.

Extremely limited power
Many devices run on a tiny battery.

Huge distances
Data may cross very large areas.

Heterogeneity
Many technologies mixed in one deployment.

Scale
From a handful of devices to millions.

Choosing the right communication technology for an application really matters for the system's success.

1

Recap: where communication lives

Unit I's five-layer architecture, with the Network Layer in focus.

5 · Business	Business-level decisions
4 · Application	User-facing apps and dashboards
3 · Processing	Stores and analyzes the data
2 · Network	**Transmits data to where it is processed**
1 · Perception	Sensors and actuators

Unit II lives in the Network Layer

Its entire job is to carry what the Perception Layer collected to wherever it is processed.

- **Communication technologies**
- **Gateways**

Almost all of Unit II is about this one layer, because there are so many competing options.

2 Communication runs both ways

Readings travel up to the cloud and users; commands travel back down to actuators.



The closed loop from Unit I: sense the world, decide what to do, then act on it.

Readings go up

Sensor data flows from the device toward the cloud and the people who use it.

Commands come down

Decisions travel back to actuators, closing the loop between sensing and acting.

Every arrow is a choice

Each hop needs a technology, a protocol and a way to be addressed.

3 Why moving data is so hard

One of the hardest, most consequential engineering problems in IoT.

Unreliable links

Data often travels over inherently unreliable wireless links.

Limited power

Devices frequently run on extremely limited battery power.

Huge distances

Data sometimes has to cross genuinely large geographic areas.

Heterogeneity

Devices from many makers, using many technologies, in one deployment.

Scale

From a handful of devices in one apartment to millions across a power grid.

So the choice matters

Picking the right technology for the application really matters for the system's success.

This is exactly why an entire unit is dedicated to communication and networking.

Power shapes every choice

A reminder from the Unit I battery example.

80 mA

while transmitting, for 2 s per reading

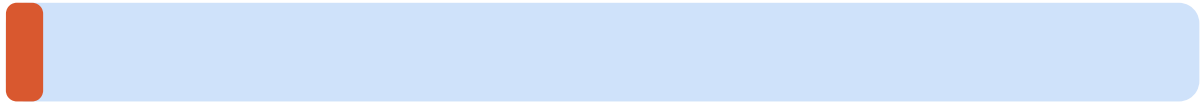
0.5 mA

while idle between readings

≈ 109 days

on a 2,000 mAh battery, one reading every 10 minutes

One 600-second cycle



2 s transmitting **598 s idle**

(transmit slice enlarged for visibility)

Idle current dominates the average because the device spends nearly all of its time idle.

The lesson to carry forward

Battery-powered IoT design obsesses over how much time the radio is active. Expect this theme again and again as we compare communication technologies.

A great link that drains the battery in weeks is a poor choice for a field sensor.

One deployment, many technologies

Heterogeneity and scale.

Heterogeneity: everything is different

On Bluetooth

A short-range link

On a cellular link

A wide-area link

Full Linux OS

A capable small computer

No OS at all

A tight loop of bare-metal code

Unlike a corporate network of similar laptops, one IoT deployment can mix all four.

Scale: two extremes

A handful

devices in one smart apartment

500 to 50,000

sensors in a citywide network

Millions

devices across a national power grid or farming region

Whatever we choose must work sensibly at both extremes and across all of these devices.

4 Four ways to look at communication

Unit II is organized around four questions.

Short-range

How do nearby devices connect?

- Bluetooth / BLE
- Zigbee
- RFID
- NFC
- Wi-Fi

Long-range

How do devices reach across distance?

- LoRaWAN
- NB-IoT
- LTE-M
- 4G/5G for IoT

Protocols

How are messages structured and exchanged?

- MQTT
- CoAP
- HTTP/HTTPS
- AMQP
- WebSockets

Networking

How does the whole network hang together?

- IPv4 and IPv6
- 6LoWPAN
- Wireless sensor networks
- Gateways
- Addressing and routing

A comparison table at the end sets them side by side: range, power, bandwidth and use-case.

5 Short-range versus long-range

Devices with wildly different power budgets and range needs all have to connect.

Short-range options

Suited to devices that sit close to each other or to a gateway.

- Bluetooth / BLE
- Zigbee
- RFID
- NFC
- Wi-Fi

Long-range options

Built to cover large distances; most are tuned for low power.

- LoRaWAN
- NB-IoT
- LTE-M
- 4G/5G for IoT

Four dimensions to compare

- Range
- Power
- Bandwidth
- Use-case

Each technology suits its own constraints; there is rarely one best answer.

6 How the pieces stack

A simple mental model for organizing Unit II.

Protocols	**Structure and exchange the actual messages.** MQTT, CoAP, HTTP/HTTPS, AMQP, WebSockets	What do the messages say?
Networking concepts	**Give every device an address and a route.** IPv4/IPv6, 6LoWPAN, gateways, WSN	How do they find their way?
Communication technologies	**Physically move the data, over short or long range.** Bluetooth, Zigbee, Wi-Fi, LoRaWAN, NB-IoT, LTE-M, 4G/5G	How do the bits travel?

A teaching model, not a strict standard. Real systems blur these boundaries, but the split keeps Unit II clear.

The Unit II roadmap

Six topics, 180 minutes.



1 Introduction

You are here

Why communication is the backbone of IoT

2 Short-range

Bluetooth/BLE, Zigbee, RFID, NFC, Wi-Fi

3 Long-range

LoRaWAN, NB-IoT, LTE-M, 4G/5G

4 Protocols

MQTT deep dive, CoAP, HTTP/HTTPS, AMQP, WebSockets

5 Networking

IPv4/IPv6, 6LoWPAN, WSN, gateways, routing

6 Comparison & recap

Side-by-side table, self-assessment, Unit III preview

You are here: topic 1, the introduction. Protocols and short-range technologies get the most time.

By the end of Unit II

You will be able to:

Choose

Pick the right communication technology for a given application.

Explain

Describe how MQTT works: publish-subscribe, brokers and topics.

Place

Say where IPv6, 6LoWPAN, gateways and wireless sensor networks fit.

Compare

Weigh technologies side by side on range, power, bandwidth and use-case.

Look at any IoT system and ask: what is likely carrying this data across the network?

The big picture

The backbone	Sensing creates the data; communication carries it, in both directions.
The Network Layer	Unit II lives here: moving data reliably to where it is processed.
The hard part	Unreliable links, tiny batteries, huge distances, variety and scale.
Four families	Short-range, long-range, protocols and networking concepts.
Up next	Topic 2: short-range communication technologies.

One-line takeaway

“Sensing creates the data, but communication carries it. Choose the right link, protocol and network, and the whole system works.”

Short-range | Long-range | Protocols | Networking | Comparison

Up next: Bluetooth, Zigbee, RFID, NFC and Wi-Fi.

Short-Range Communication

Five technologies connect nearby devices: Bluetooth LE, Zigbee, RFID, NFC and Wi-Fi. Each trades range, power and speed differently.



Nearby devices, connected.

In Simple Words

Short-range links cover a few centimeters up to a building. They are how most devices first reach a network.

Bluetooth LE	tens of meters
Zigbee	tens of meters per hop
RFID	centimeters to several meters
NFC	a few centimeters
Wi-Fi	tens of meters, indoors

What every choice trades off

Comparing short-range technologies always comes back to five questions:

Range
How far does the link reach?

Power
Battery for years, or a wall socket?

Data rate
A trickle of readings, or a video stream?

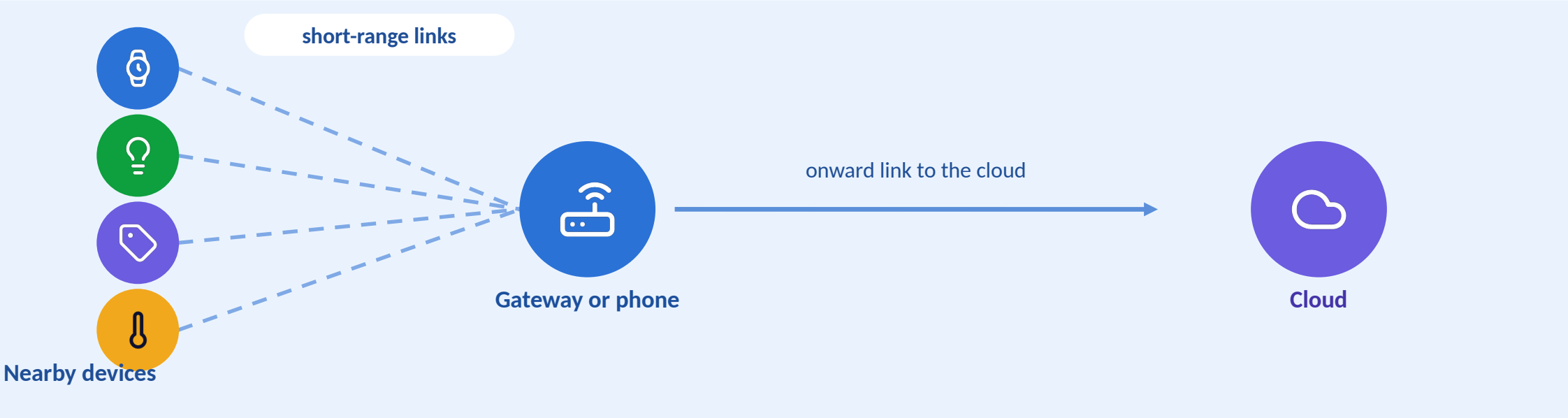
Topology
Point-to-point, star or mesh?

Ecosystem
Which phones, hubs and routers already exist?

No single technology wins everywhere: match the link to the application.

1 Where short-range links fit

The last few meters between a device and the rest of the network.



Cover a small area

From a few centimeters to a room or a building.

Built to sip power

Several are designed to run a long time on a small battery.

Talk to a nearby hub

Devices usually reach a gateway or phone, which forwards data onward.

When the distance is too large, long-range technologies (Topic 3) take over.

Bluetooth and Bluetooth Low Energy

2.4 GHz
license-free ISM band

Tens of meters
typical range

Up to 2 Mbps
BLE data rate

Bluetooth Classic

Built for continuous streams, such as audio.

Typical uses

- Headsets
- Speakers

Bluetooth Low Energy (BLE)

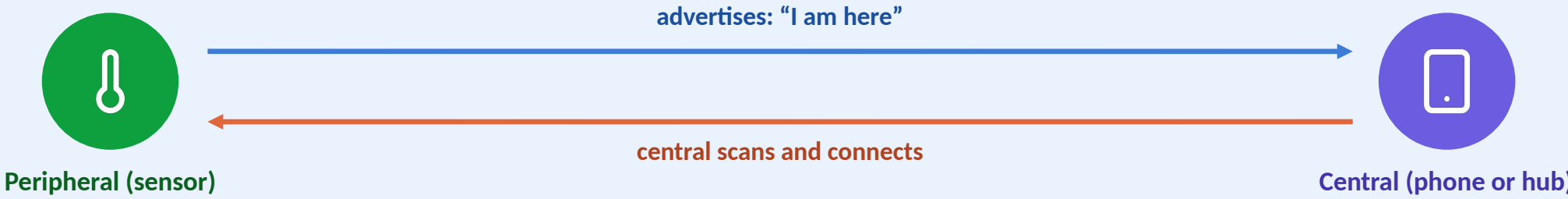
Built for short bursts of small data at very low power.

Typical uses

- Wearables
- Beacons
- Smart locks
- Sensors

For IoT, BLE is usually the one that matters: small data, long battery life.

BLE: how a connection works



1 Advertise

A peripheral broadcasts tiny packets so nearby devices can discover it.

2 Connect

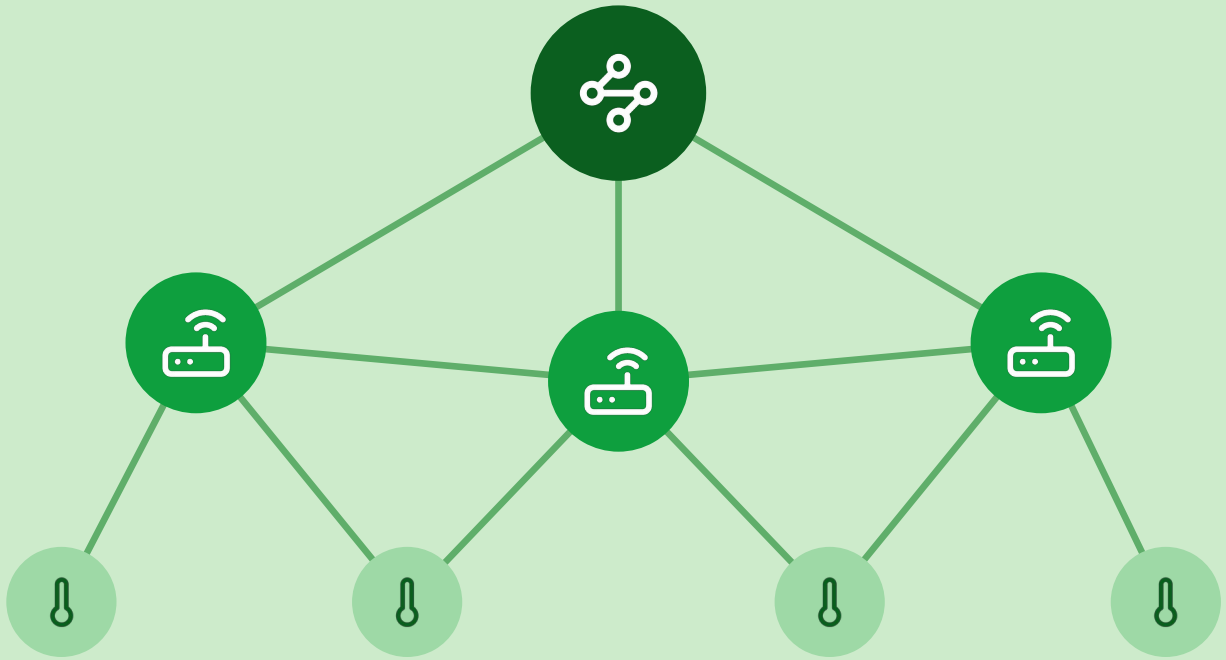
A central device scans, picks a peripheral and opens a connection.

3 Exchange

Data lives in characteristics grouped into services. The central reads, writes or subscribes to notifications, then both sleep to save power.

For bigger installations such as lighting, Bluetooth also supports mesh networking.

Zigbee: a low-power mesh



IEEE 802.15.4 | 2.4 GHz | Up to 250 kbps

Range hop by hop

Each router extends coverage: tens of meters per hop, farther across a mesh.

Self-healing

Messages route around a failed or blocked node.

Built to scale

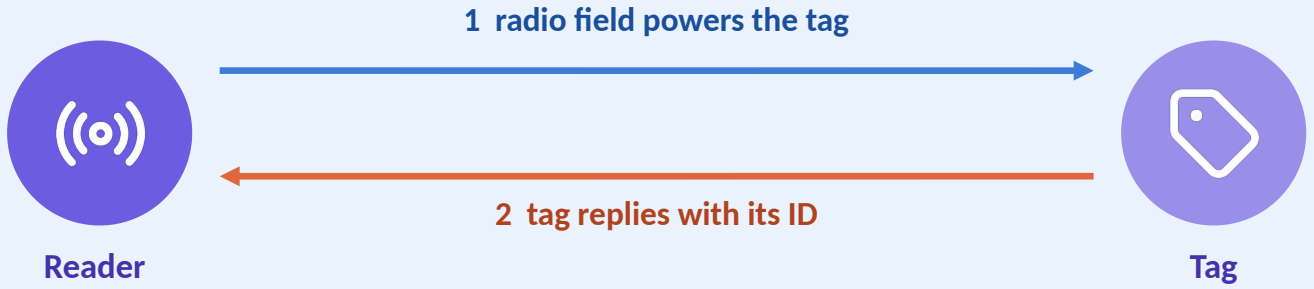
Low data rate and low power suit many small nodes.

Common in smart lighting and home automation, industrial monitoring and smart metering.

RFID: recap and communication angle

Recap from Unit I

- A reader wirelessly reads a unique ID from a tag.
- No line of sight is needed.
- Passive tags have no battery: they draw power from the reader's signal.



The frequency band sets the range

LF about 125 kHz

Read range of centimeters

HF 13.56 MHz

Up to about a meter

UHF 860 to 960 MHz

Several meters

A tag mostly answers "who am I?", ideal for asset tracking and access badges.

NFC: recap and communication angle

RFID's short-range cousin, built for taps.

13.56 MHz
same band as HF RFID

A few cm
range, at most

Up to 424 kbps
data rate

Reader / writer

A phone reads or writes a passive tag, such as a poster, label or smart card.

Peer-to-peer

Two devices exchange small amounts of data, often to start a Bluetooth or Wi-Fi pairing.

Card emulation

A phone behaves like a contactless payment or access card.

The tiny range is a feature: you must be right next to the reader, which keeps payments secure.

Wi-Fi: fast, everywhere, power-hungry

IEEE 802.11
the standards family

2.4 • 5 • 6 GHz
bands in use

Mbps to Gbps
data rate

Strengths

Fast	Handles video, firmware updates and heavy data.
Everywhere	Uses access points already in homes and offices.
IP-native	Devices join the internet directly.

Trade-offs

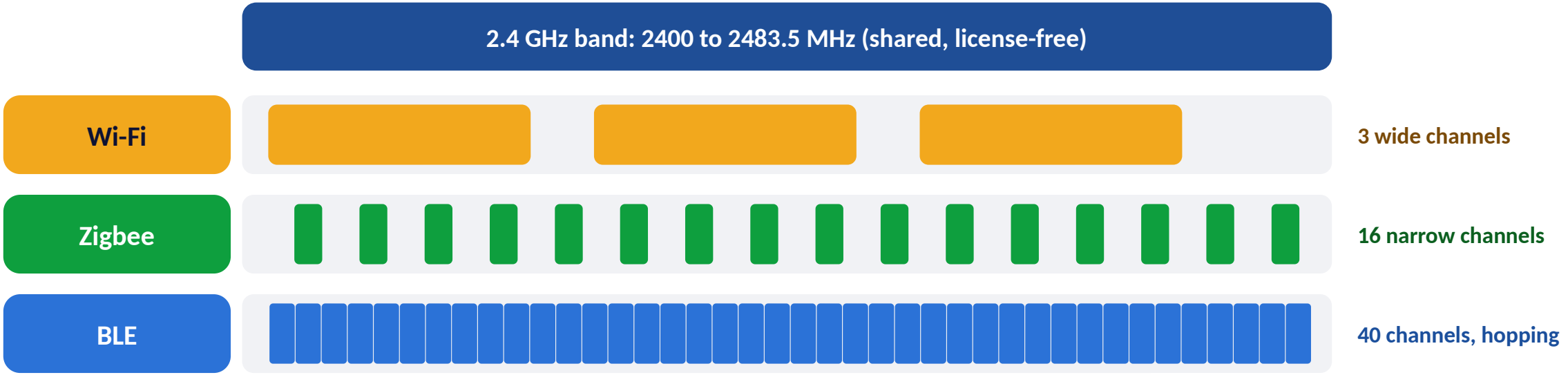
Power-hungry	Tough on small batteries.
Limited range	Tens of meters indoors, weakened by walls.
Crowded	Shares the 2.4 GHz band with many devices.

Recall the ESP32 from Unit I: Wi-Fi and Bluetooth built into a single chip.

2

Three of the five share one crowded band

Wi-Fi, Zigbee and Bluetooth LE all live in the 2.4 GHz ISM band.



Simplified: Wi-Fi channels 1, 6 and 11; Zigbee channels 11 to 26; BLE channels 0 to 39.

The consequence

Overlapping transmissions cause interference, dropped messages and shorter effective range.

The countermeasures

Zigbee and Wi-Fi pick a clear channel; BLE hops between channels to dodge noise.

Other bands

RFID and NFC use different frequencies, such as 13.56 MHz, so they do not join this crowd.

A link that fights interference is an unreliable link, the first problem from Topic 1.

3 Which one should I use?

A first-pass guide from need to technology.

If your need is...	Technology	Why
Tap to pay, pair or check in	NFC	A few centimeters is a security feature.
Track tagged assets or badges	RFID	Tags are tiny and passive ones need no battery.
A wearable or sensor talking to a phone	Bluetooth LE	Small bursts of data at very low power.
Many low-power nodes across a building	Zigbee	Mesh hops extend coverage.
Heavy data, or an existing network	Wi-Fi	High speed, with the access points already there.

A starting point, not a rule: real projects often combine several.

4 Short-range technologies side by side

Typical values only: always check the datasheet for your exact part.

Technology	Band	Typical range	Data rate	Power	Topology
Bluetooth LE	2.4 GHz	Tens of meters	Up to 2 Mbps	Very low	Point-to-point, broadcast, mesh
Zigbee	2.4 GHz (also sub-GHz)	Tens of meters per hop	Up to 250 kbps	Low	Mesh
RFID	LF, HF or UHF	Centimeters to several meters	Low (ID reads)	Passive tags need no battery	Reader to tag
NFC	13.56 MHz	A few centimeters	Up to 424 kbps	Very low	Tap: peer-to-peer or tag
Wi-Fi	2.4 / 5 / 6 GHz	Tens of meters indoors	Mbps to Gbps	Higher	Star via access point

A fuller comparison, with long-range and protocols, closes Unit II (Topic 6).

The big picture

Bluetooth LE	Small bursts of data at very low power: the phone-friendly choice.
Zigbee	A low-power, self-healing mesh that hops across a building.
RFID	Readers identify tags, often battery-free, over centimeters to meters.
NFC	Taps over a few centimeters: payments, access and quick pairing.
Wi-Fi	Fast and everywhere, but hungry for power.

One-line takeaway

“Short-range technologies get devices onto the network. Pick the one whose range, power and speed fit the job.”

Bluetooth LE | Zigbee | RFID | NFC | Wi-Fi

Up next: long-range technologies: LoRaWAN, NB-IoT, LTE-M and 4G/5G.

Going the Distance in IoT

Short-range links end at a room or a building. Long-range links carry a small reading across kilometres to a tower or a gateway, then on to the cloud.

Small data. Long distance.

In Simple Words

A soil sensor in a distant field cannot reach your Wi-Fi. A long-range link carries its tiny reading for kilometres, on one battery, for years.

Field device
Senses and sends a small reading

Tower or gateway
Picks up the signal kilometres away

Cloud
Stores, analyzes and shows it

What do we ask of it?

Long-range IoT links are designed around five needs:

Kilometres of range
From a farm or a street to a tower.

Years on a battery
The radio sleeps almost all the time.

Tiny messages
A reading, an alert or a location.

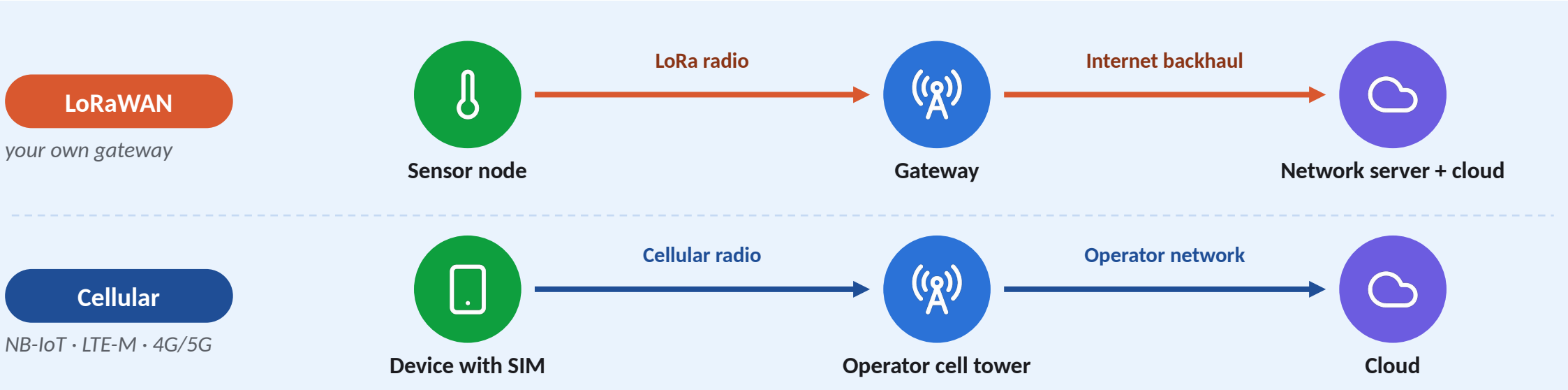
Low cost per device
Cheap modules and simple hardware.

Massive scale
Thousands of devices per gateway or cell.

Long range usually means trading speed for reach and battery life.

1 Two ways to go the distance

Topic 2 connected nearby devices. Now the data must travel kilometres.



You own the last mile

LoRaWAN: place your own (or a community) gateway. No SIM card in the sensor.

The operator owns it

Cellular: towers and spectrum belong to a mobile operator, and every device carries a SIM.

Same destination

Both paths end in the cloud, where the data pipeline from Unit I takes over.

Ask early: do we run the network ourselves, or rent an operator's?

The Topic 3 roadmap

Six stops, 35 minutes.



1 Foundations

LPWAN idea, spectrum and the speed ladder

5 min

2 LoRaWAN

Architecture, spreading factor and device classes

9 min

3 NB-IoT

Cellular LPWAN: deep coverage, PSM and eDRX

6 min

4 LTE-M

Cellular LPWAN that moves, and how it differs from NB-IoT

5 min

5 4G/5G for IoT

LTE categories, 5G service types and RedCap

5 min

6 Compare & choose

Side-by-side table, decision guide and scenarios

5 min

Topic 3 of 6 in Unit II. Next, Topic 4 moves from radios to protocols such as MQTT.

By the end of this topic

You will be able to:

Explain

What an LPWAN is, and why range, battery life and data rate pull against each other.

Describe

How a LoRaWAN network fits together: devices, gateways, network server and spreading factor.

Contrast

The two cellular LPWAN options, NB-IoT and LTE-M, and where 4G and 5G fit in.

Choose

The right long-range technology for an application, using range, power, bandwidth and use-case.

Ask four questions of any deployment: how far, how much data, how often, and does it move?

2

What is an LPWAN?

Foundations

Low-Power Wide-Area Network: small data, sent far, on a tiny battery.

Kilometres

Wide-area range

Reach from a few km in towns to 10+ km in open country.

Years

Low power

The radio sleeps almost all the time. Batteries are designed to last up to about ten years.

Kilobits

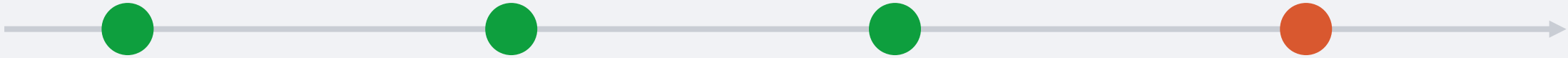
Low data rate

Bytes, not megabytes: a reading, an alert, a location.

Where LPWAN sits on the range ladder

Topic 2: short-range

Topic 3: long-range



NFC, RFID	BLE, Zigbee	Wi-Fi	LoRaWAN, NB-IoT, LTE-M, 4G/5G
centimetres to a metre	tens of metres	around a building	kilometres

LPWAN gives up speed and low latency to gain reach and battery life.

3

Licensed vs unlicensed spectrum

Foundations

The first big fork in the road for long-range IoT.

Unlicensed spectrum

LoRaWAN

- Free to use within power and airtime limits
- Sub-GHz ISM band: 865–867 MHz in India
- Deploy your own gateways, no SIM needed
- Shared band: interference, no guaranteed quality

Licensed spectrum

NB-IoT · LTE-M · 4G/5G

- The operator holds exclusive rights to the band
- Runs on the mobile network; every device has a SIM
- Better reliability and quality of service
- Subscription cost; coverage depends on the operator

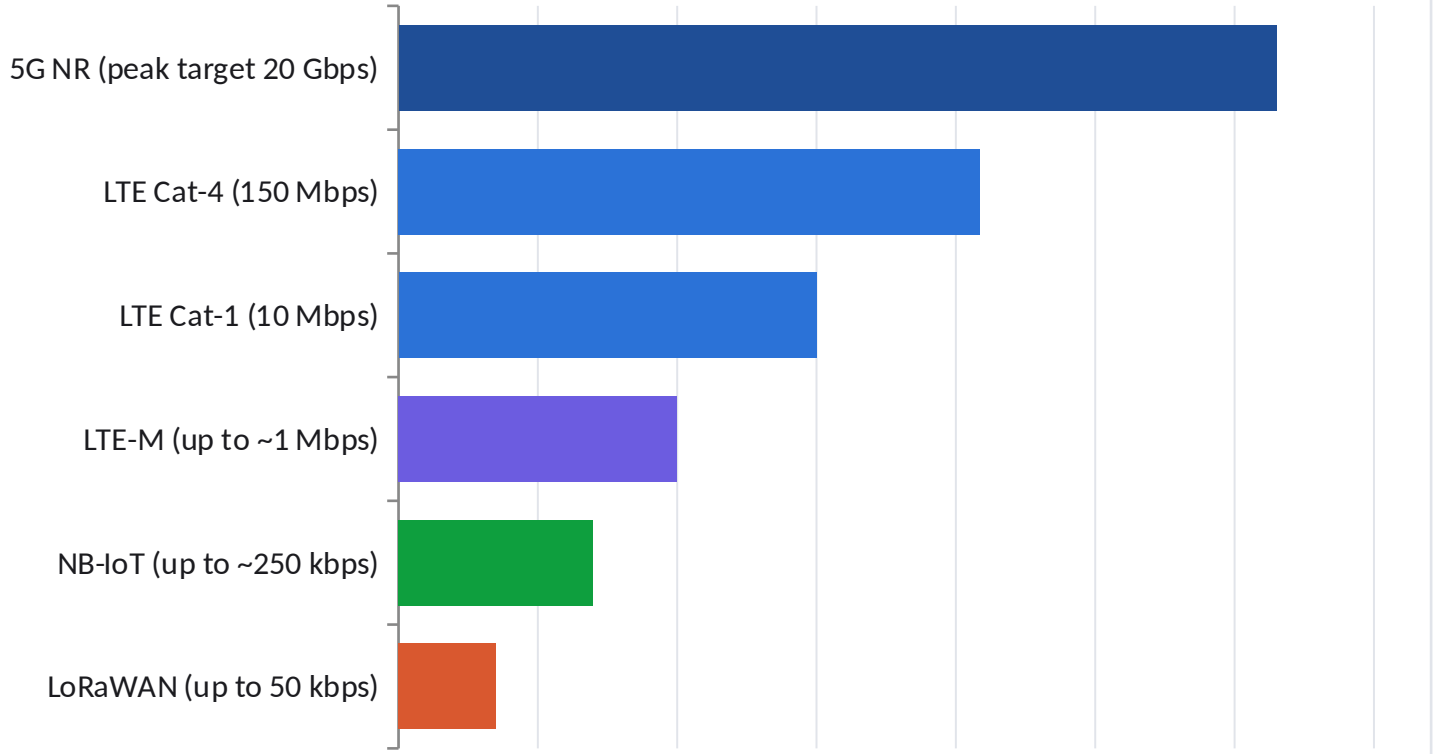
Licensed spectrum buys reliability and coverage; unlicensed buys freedom and low cost.

4

The speed ladder

Foundations

Each rung up costs power, money or both.



More speed, more power

Radios that move more bits stay awake longer and draw more current.

More speed, more cost

Faster modules and richer data plans cost more per device.

Most sensors need little

A soil reading is a few bytes. Even 1 kbps is plenty.

Match the speed to the message: a soil reading needs bytes, a video stream needs megabits.

5

LoRa and LoRaWAN

Two layers, two jobs.

LoRaWAN

Application	**Your data and the apps that use it** Dashboards, alerts and cloud services fed by the application server.	What do we do with the data?
LoRaWAN	**The network protocol (MAC layer)** Addressing, AES-128 security, device classes and adaptive data rate. An open standard maintained by the LoRa Alliance.	Who talks to whom, and how safely?
LoRa	**The radio (physical layer)** Chirp spread spectrum radio from Semtech on a sub-GHz band. Trades data rate for range and sensitivity.	How do the bits travel?

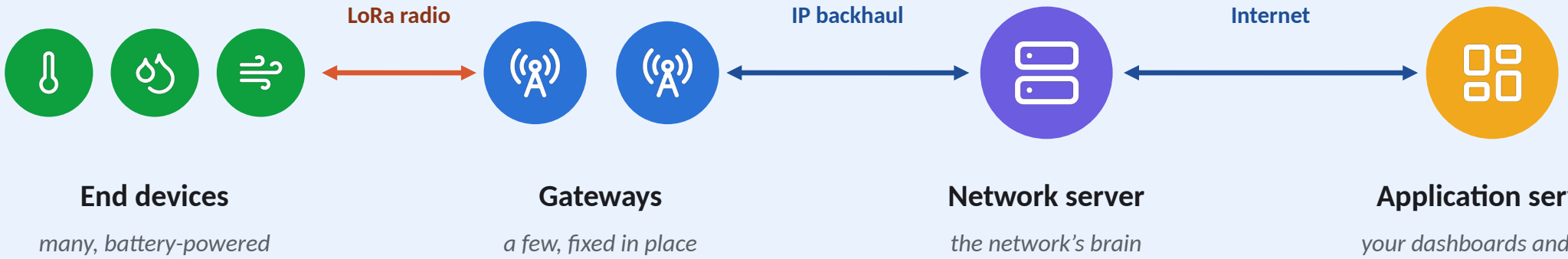
LoRa is the road; LoRaWAN is the rulebook that says who may drive, when, and how safely.

6

How a LoRaWAN network is built

LoRaWAN

A star-of-stars: many devices, a few gateways, one network server.



End devices

Sensors with a LoRa radio. They just transmit, with no pairing to a single gateway.

Gateways

Hear every device in range and forward the raw packets to the server over IP.

Network server

Removes duplicates, checks security, adapts data rates and routes messages.

Application server

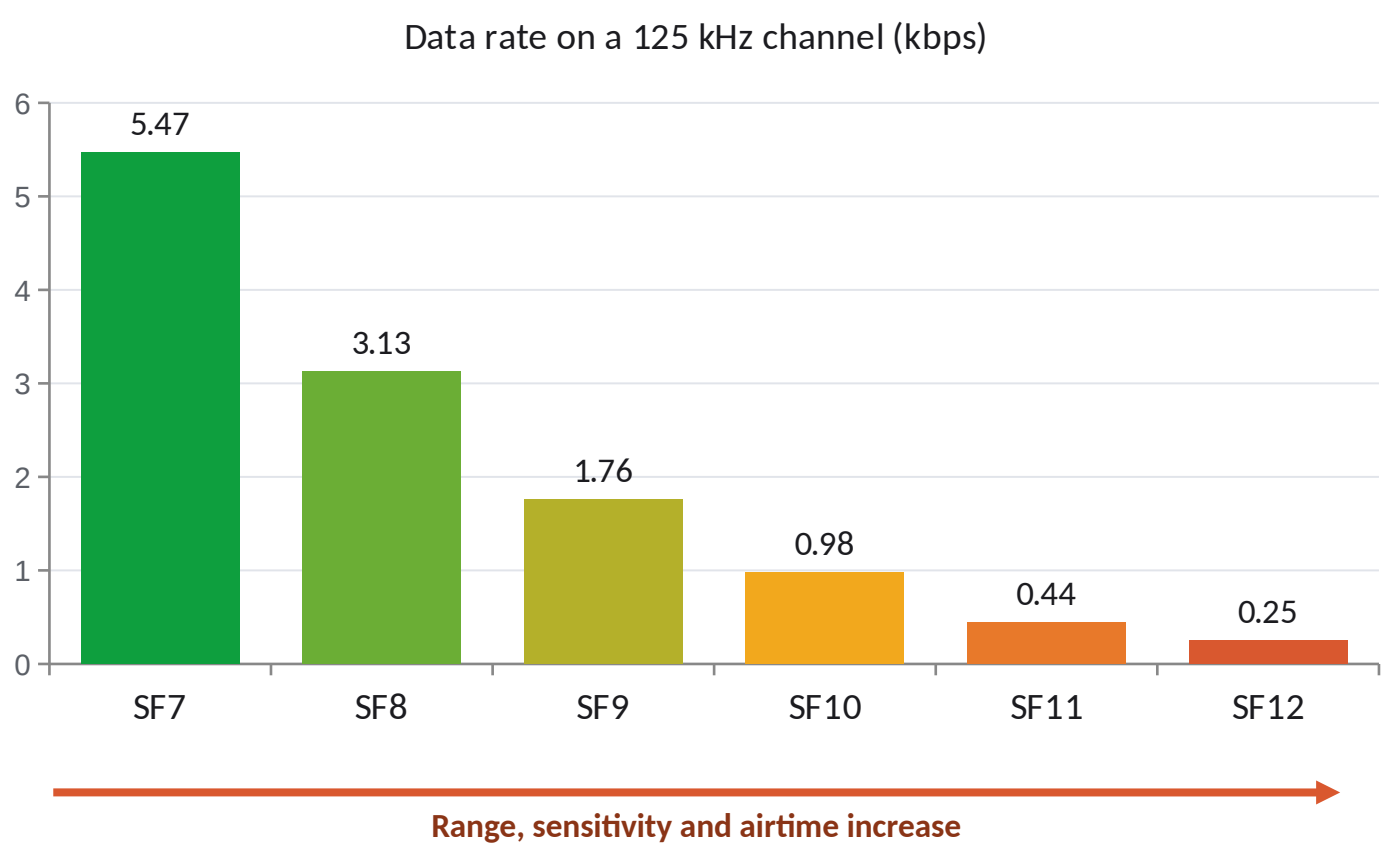
Decrypts the payload and feeds dashboards, alerts and cloud services.

A device is not tied to one gateway: every gateway in range hears it and forwards the message.

7 Trading speed for range

LoRaWAN

LoRa's spreading factor (SF7 to SF12) sets how fast or how far a message travels.



A higher spreading factor means

- Longer range, better reception
- Slower data rate
- Roughly double the airtime per step
- More energy per message

Adaptive Data Rate

The network server picks the lowest SF that still works. Close devices go fast and save battery; distant ones slow down to be heard.

Rule of thumb: use the lowest spreading factor that still reaches the gateway.

8

Three device classes: A, B, C

LoRaWAN

Classes trade listening time against power and responsiveness.

A Class A

Every device supports it

How it listens

Sends when it has data, then opens two short receive windows. Sleeps the rest of the time.

Power use

Best for

Battery sensors: soil, water, temperature.

B Class B

Scheduled listening

How it listens

Adds extra receive slots at scheduled times, kept in step by beacons from the gateway.

Power use

Best for

Devices that need commands at predictable times, such as valves.

C Class C

Always listening

How it listens

Keeps its receiver open almost all the time, except when transmitting.

Power use

Best for

Mains-powered actuators, such as street lights.

Every LoRaWAN device supports Class A; B and C buy faster downlink at a cost in power.

LoRaWAN at a glance

Key facts to remember.

2–5 km
typical range in towns and cities

15 km+
possible in open country

0.25–50
kbps data rate, small payloads

~10 years
battery-life design target

Strengths

- Long range on a sub-GHz radio
- Multi-year battery life
- Own or community gateways, no SIM
- AES-128 encryption, network and application level

Watch-outs

- Low data rate and small payloads
- Shared band: interference and airtime limits
- No latency guarantee; limited downlink
- You must run (or find) a gateway

LoRaWAN trades speed and guarantees for reach, battery life and low cost.

9

LoRaWAN in the field

LoRaWAN

One gateway can cover a whole farm; the network tunes each sensor.

Near sensors: SF7–SF8, fast

Far sensors: SF10–SF12, slow



Where LoRaWAN shines

Smart agriculture
Soil moisture, weather, irrigation

Smart metering
Water and gas meters, read remotely

Smart city
Parking, street lights, waste bins

Asset and animal tracking
Low-power location on a farm or campus

Environmental monitoring
Air quality, river levels, pollution

Networks can be private, or shared through community projects such as The Things Network.

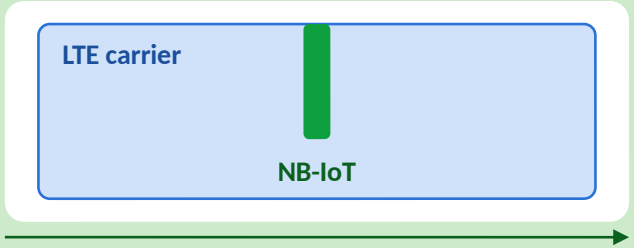
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NB-IoT: an LPWAN inside LTE

NB-IoT

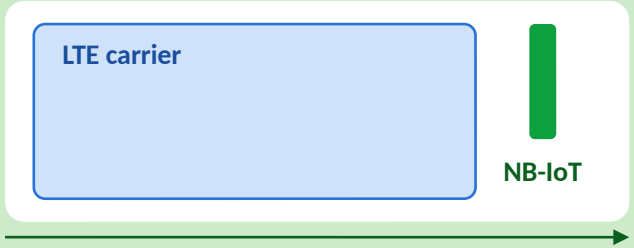
Narrowband IoT: 3GPP Release 13, licensed spectrum, run by the operator.

In-band



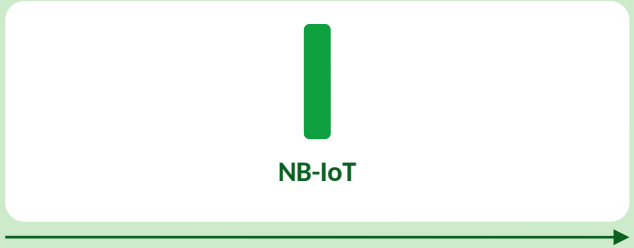
Inside an LTE carrier, using one 180 kHz resource block.

Guard-band



In the unused guard band at the edge of an LTE carrier.

Stand-alone



In its own spectrum, for example re-farmed from GSM.

180 kHz

one narrow slice, about one LTE resource block

Release 13

3GPP standard; also counts as 5G massive IoT

SIM + operator

licensed spectrum and the operator's network

Often added to existing LTE towers as a software upgrade, so coverage can spread quickly.

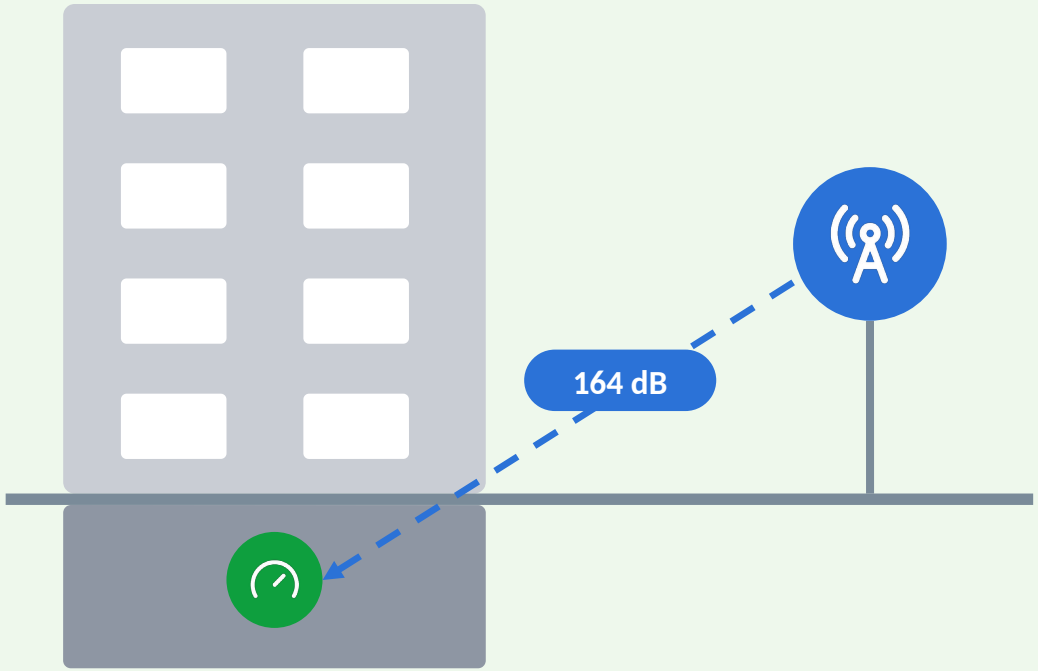
11

Deep coverage, long battery

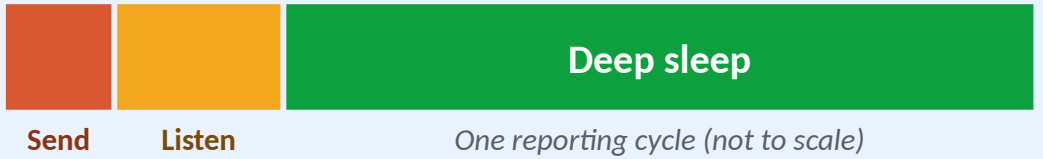
NB-IoT

Two headline strengths: reach into basements, and years of sleep.

164 dB: reaching the basement



Sleeping to save power



PSM: Power Saving Mode

The device sleeps deeply and cannot be reached until it wakes to send. Ideal for meters that only report.

eDRX: extended DRX

The device sleeps between listening windows that can be many minutes apart, so it can still receive commands.

Design target: more than ten years on a 5 Wh battery, thanks to deep sleep.

12

NB-IoT: strengths and limits

NB-IoT

Superb for fixed, quiet devices. Not for anything that moves or talks a lot.

Strengths

- **Deep coverage**
  basements, underground, inside concrete
- **Ten-year battery**
  deep sleep with PSM and eDRX
- **Low-cost modules**
  simple, narrowband radio
- **Massive scale**
  tens of thousands of devices per cell

Limits

- **Low speed**
  tens to a few hundred kbps at best
- **High latency**
  seconds, not milliseconds
- **Limited mobility**
  no seamless handover; best for fixed devices
- **Needs an operator**
  NB-IoT coverage and a subscription

Typical uses

Water and gas meters | Smart parking | Waste bins | Street lighting | Fixed sensors

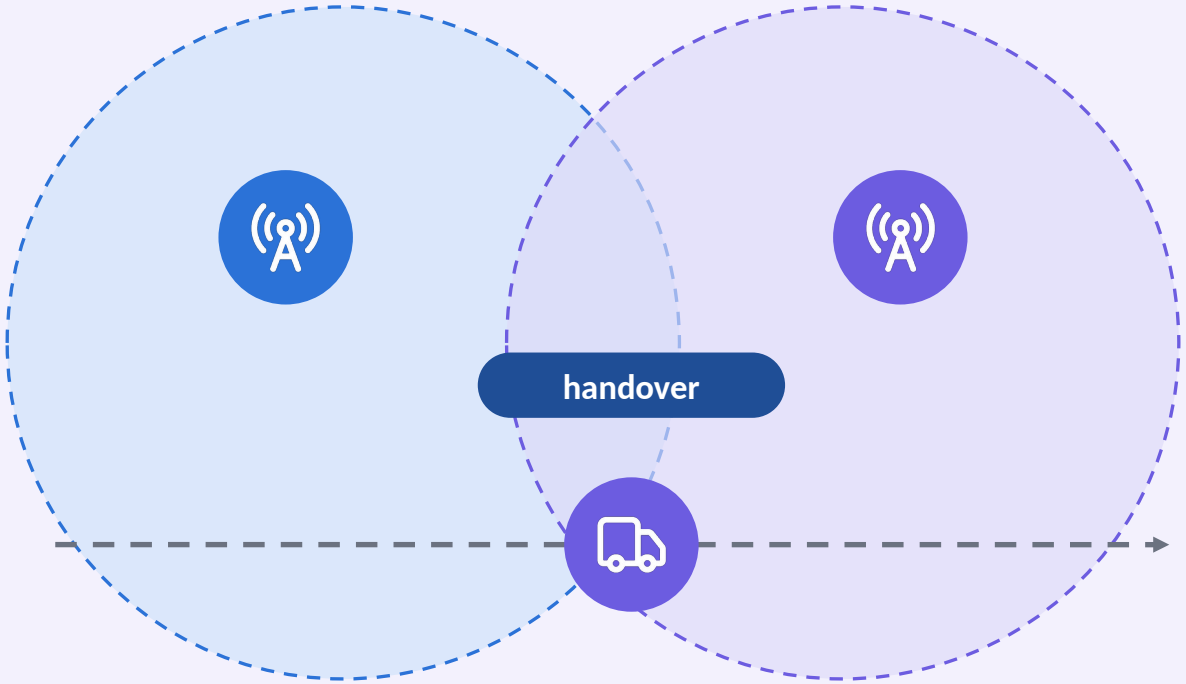
Choose NB-IoT for devices that sit still, send little, and hide in hard-to-reach places.

13

LTE-M: cellular that can move

LTE

LTE Cat-M1, also called eMTC: 3GPP Release 13 on a 1.4 MHz channel.



The session continues as the device moves between cells

~1 Mbps

on a 1.4 MHz channel

Handover

keeps the session alive on the move

Voice

VoLTE support for wearables and alarms

OTA updates

fast enough to update firmware over the air

LTE-M keeps NB-IoT's power-saving modes but adds mobility, voice and more speed.

14

NB-IoT vs LTE-M

LTE-M

Two cellular LPWAN options: same family, different strengths.

	NB-IoT	LTE-M
Channel bandwidth	180 kHz	1.4 MHz
Peak data rate	Tens to a few hundred kbps	Up to about 1 Mbps
Mobility	Fixed; no seamless handover	Full handover, works on the move
Voice	No	Yes (VoLTE)
Latency	Seconds	Much lower
Coverage (link budget)	About 164 dB, deepest reach	About 156–161 dB, still strong
Best for	Fixed meters and sensors	Tracking, wearables, alarms

Both use licensed LTE spectrum and a SIM, and both use PSM and eDRX to sleep.

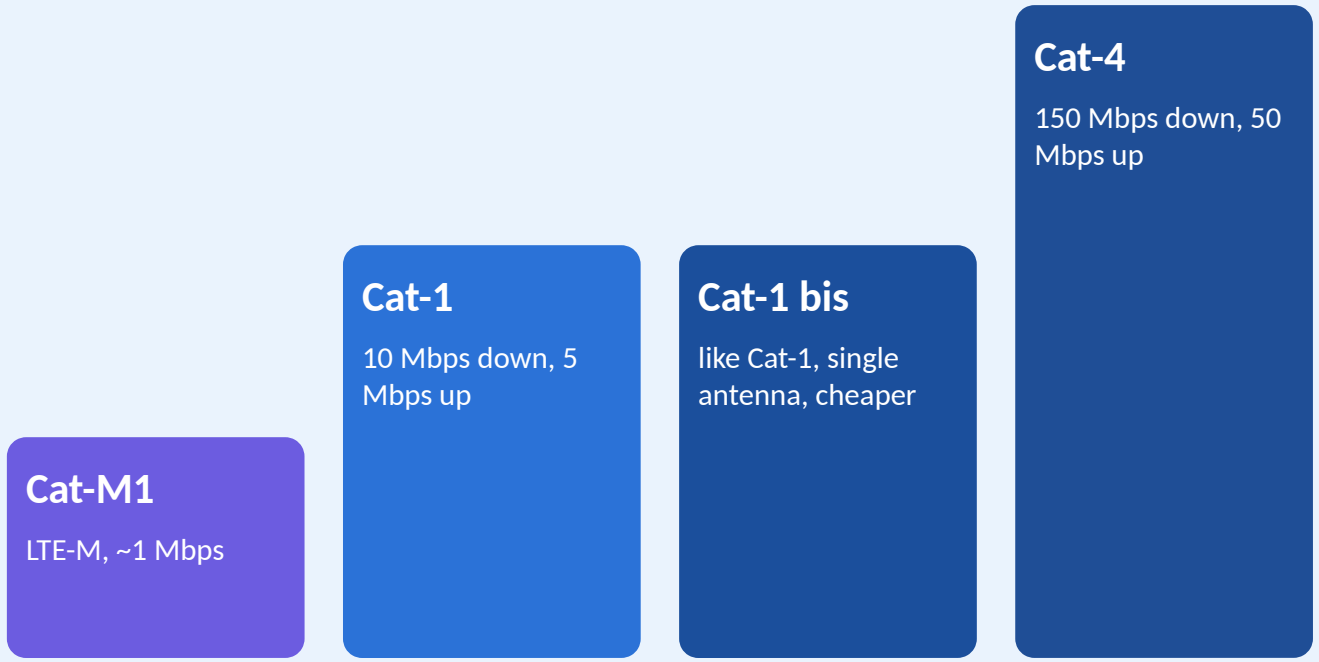
15

4G for IoT: real bandwidth

4G/5G

When a device needs megabits, it climbs the LTE category ladder.

Climbing the ladder: more speed, more power, more cost



Not to scale

Where it fits

- **Vehicle telematics and dash cameras**
- **Payment terminals and kiosks**
- **Gateway backhaul for LoRaWAN and other networks**

Why now?

As 2G and 3G networks close in many countries, devices move to LTE-based options.

Pay for bandwidth only when the job needs it: it costs battery life and money.

16

5G for IoT: three service types

4G/5G

One radio standard, three very different promises.

eMBB

Enhanced mobile broadband

20 Gbps

peak-rate target

HD video, AR/VR and cameras.

URLLC

Ultra-reliable, low latency

≈ 1 ms

latency target

Factory robots, remote control, autonomous vehicles.

mMTC

Massive machine-type communication

1 million

devices per km²

Sensor swarms. NB-IoT and LTE-M count as 5G mMTC.

5G RedCap (Release 17)

A middle path between full 5G and LPWAN: lower cost, size and power than full 5G, with peak rates below 250 Mbps.

- Wearables
- Video cameras
- Industrial sensors

5G is a toolbox: pick the service type that matches the device, not the biggest number.

A first look: side by side

Range, power, bandwidth and use-case.

	LoRaWAN	NB-IoT	LTE-M	4G/5G
Spectrum	Unlicensed sub-GHz (865–867 MHz in India)	Licensed cellular	Licensed cellular	Licensed cellular
Range	2–5 km urban, 15 km+ rural	Cell coverage, deep indoors	Cell coverage	Cell coverage
Power	Up to ~10 years on a battery	10+ years (design target)	Years on a battery	Highest draw; often mains-powered
Bandwidth	≈ 0.25–50 kbps	≈ 20–250 kbps	Up to ~1 Mbps	10 Mbps to Gbps
Mobility	Limited	Fixed	Yes, with handover	Yes
Best use-case	Farms, campuses, metering	Fixed meters, parking	Tracking, wearables	Video, vehicles, factories

No single winner. The full comparison closes Unit II.

17

How to choose

A quick decision guide, in four questions.

1 Does it need video, or megabits of data?

Yes → **4G / 5G**
Real bandwidth. Accept higher power and cost.

No ↓

2 Does it move, need voice, or need frequent firmware updates?

Yes → **LTE-M**
Handover, voice and ~1 Mbps, still low power.

No ↓

3 Is it fixed, sends little, and sits within operator coverage (even underground)?

Yes → **NB-IoT**
Deep coverage and the longest battery life.

No ↓

4 Is it remote, outside operator coverage, or should you own the network?

Yes → **LoRaWAN**
Your own gateways, no SIM, very long range.

A teaching shortcut: real projects also weigh coverage maps, cost, regulation and contracts.

Which one would you pick?

Decide first, then read the answer.

1 Water meters in city basements, read once a day.

NB-IoT — Fixed, tiny data, deep coverage.

2 Delivery trucks reporting their location across the state.

LTE-M — It moves, so it needs handover.

3 Soil sensors on a remote farm with no mobile signal.

LoRaWAN — Own gateway, huge range, years of battery.

4 A traffic camera streaming live video to a control room.

4G / 5G — It needs megabits of bandwidth.

Ask: how far, how much data, how often, and does it move?

The big picture

The LPWAN idea	Small data, long distance, tiny battery. Speed is the price.
LoRaWAN	Unlicensed sub-GHz radio, your own gateways, spreading factor and device classes.
NB-IoT	Cellular and narrowband: deep coverage and the longest battery life for fixed devices.
LTE-M	Cellular with mobility, voice and about 1 Mbps, still low power.
4G / 5G	Real bandwidth and low latency when the job truly needs them.
Up next	Topic 4: IoT communication protocols, starting with MQTT.

One-line takeaway

“Long range is a trade: give up speed, gain distance and battery. Choose the link that fits the message, not the biggest pipe.”

LoRaWAN | NB-IoT | LTE-M | 4G/5G | Comparison

Up next: IoT communication protocols. MQTT deep dive, CoAP, HTTP/HTTPS, AMQP and WebSockets.

Giving Bits a Meaning

A radio delivers bits. A protocol tells the receiver what those bits are, who they are for, and whether they arrived.

One language. Many devices.

In Simple Words

A soil sensor can reach the internet over Wi-Fi or LoRaWAN. But the server still needs to know: is this a temperature or a moisture value, which sensor sent it, and did the message arrive? A protocol answers those questions.

Field device
Senses and sends a small reading

↓ *protocol*

Broker or server
Receives it and passes it on

↓

Cloud and apps
Store, analyze and show it

What do we ask of it?

IoT protocols are chosen around five needs:

Lightweight	Small headers, small memory, small code.
Low power	Fewer bytes on the air means a longer battery life.
Reliable delivery	Some messages must arrive; others may be lost.
Scalable	Thousands of devices, many senders and receivers.
Secure	Encrypted and authenticated, usually with TLS.

Protocol design is a trade: lightness, reliability and features pull against each other.

1

Where protocols sit

Big picture

Topics 2 and 3 were the bottom layer. Topic 4 is the top one.

Application

Topic 4 (now)

MQTT · CoAP · HTTP/HTTPS · AMQP · WebSocket

Transport

Topic 4 (now)

TCP · UDP

Network

Topic 5

IPv4 · IPv6 · 6LoWPAN

Link and radio

Topics 2 and 3

BLE · Zigbee · Wi-Fi · LoRaWAN · NB-IoT · LTE-M · 5G

Same radio, different protocol

An ESP32 on Wi-Fi can speak MQTT, CoAP or HTTP. The radio does not choose the protocol.

Same protocol, different radio

MQTT runs the same over Wi-Fi, Ethernet or a cellular SIM. Only the link below changes.

Protocols need an IP path

Everything in this topic rides on IP. Topic 5 shows how devices get an address and a route.

Ask early: is my problem the radio, the network, or the message?

The Topic 4 roadmap

Six stops, 45 minutes.



1 Foundations

Protocol stack, messaging patterns, TCP vs UDP and overhead

5 min

2 MQTT

Publish-subscribe, broker, topics, QoS and sessions

15 min

3 CoAP

REST for constrained devices, over UDP

6 min

4 HTTP / HTTPS

The web's protocol, and what it costs in IoT

5 min

5 AMQP & WebSockets

Queued back-end messaging, and a two-way web pipe

8 min

6 Compare & choose

Side-by-side table, decision guide and scenarios

6 min

Topic 4 of 6 in Unit II. Next, Topic 5 moves from messages to networking: IPv6, 6LoWPAN and gateways.

By the end of this topic

You will be able to:

Explain

The publish-subscribe and request-response patterns, and why IoT often prefers publish-subscribe.

Describe

How MQTT works: broker, topics and wildcards, the three QoS levels, retained messages and Last Will.

Contrast

CoAP, HTTP/HTTPS, AMQP and WebSockets by transport, overhead, delivery guarantees and typical use.

Choose

The right protocol for an application, using pattern, device constraints and reliability needs.

Ask four questions of any IoT conversation: who talks to whom, how often, how reliably, and on what kind of device?

2

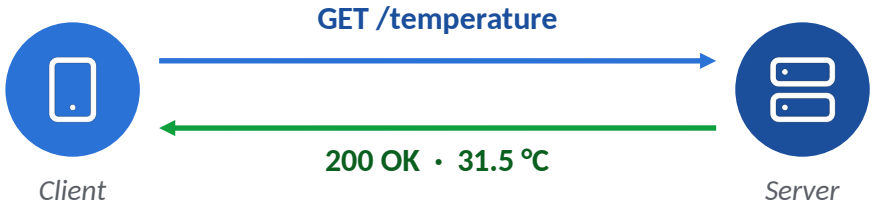
Two messaging patterns

Ask and answer, or announce and listen.

Foundations

Request / response

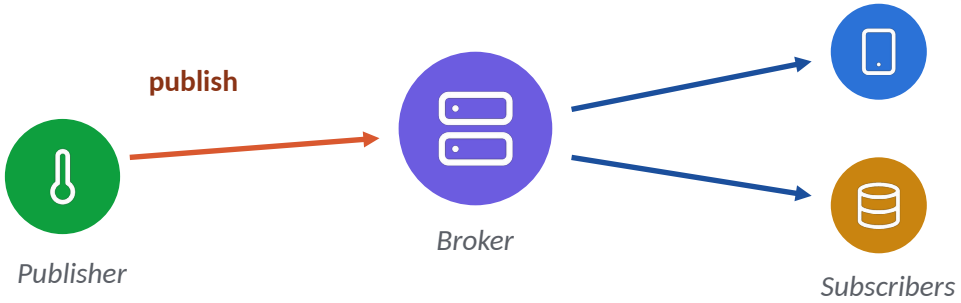
HTTP CoAP



- The client asks; the server answers
- Both sides must be online right now
- Simple and familiar, but updates need polling

Publish / subscribe

MQTT AMQP



- Publishers and subscribers never meet
- A broker routes each message by topic
- One message can reach many receivers

WebSockets is neither: a persistent two-way channel that can carry whichever pattern you build on it.

3

TCP or UDP underneath?

Foundations

The transport choice shapes everything above it.

TCP: reliable stream

MQTT · HTTP/S · AMQP · WebSocket

- Handshake first, then data flows
- Guaranteed, in-order delivery
- More overhead; the device keeps connection state
- Outbound connections pass most firewalls

vs

UDP: light datagrams

CoAP

- No connection: just send the packet
- No guarantee: the protocol adds its own
- Lightest option for tiny, sleepy devices
- Supports multicast to many devices at once

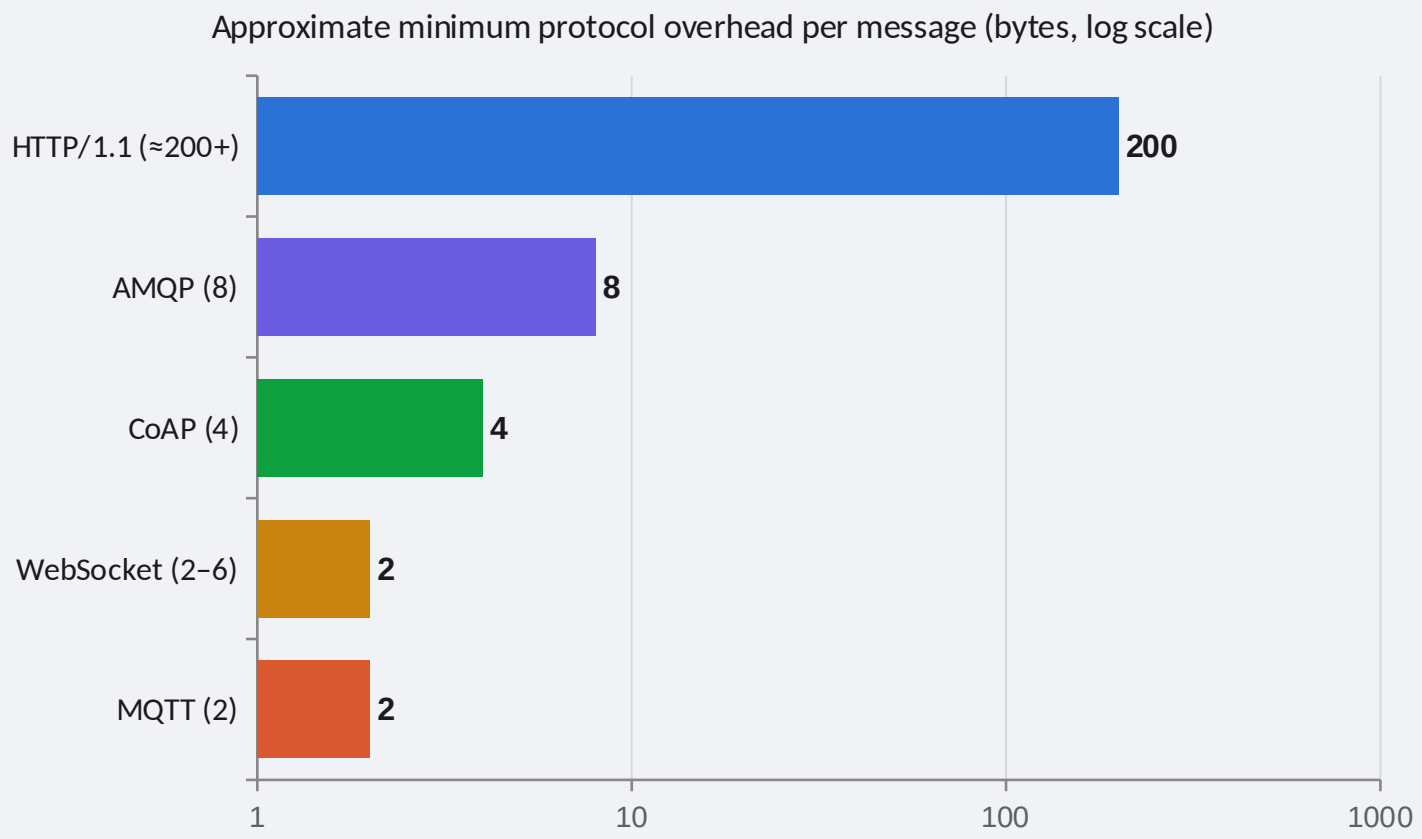
TCP buys reliability with overhead; UDP buys lightness and leaves reliability to the protocol.

4

The overhead ladder

Foundations

Every byte of wrapping costs air time, and air time costs battery.



Small headers save energy

Fewer bytes on the air means the radio is on for less time, and that is what drains a battery.

Binary vs text

MQTT, CoAP and AMQP pack fields into bytes. HTTP sends readable text headers on every request.

Read the shape, not the number

Real overhead also includes topic names, options and TLS. Compare the pattern, not exact bytes.

Match the protocol to the message: a 4-byte reading should not travel in a 200-byte wrapper.

5

What is MQTT?

MQTT

Message Queuing Telemetry Transport: lightweight publish-subscribe messaging.

2 bytes

Smallest header

The fixed header of an MQTT packet can be as small as two bytes.

1883

Default TCP port

8883 when the connection is secured with TLS.

3 levels

Delivery guarantees

QoS 0, 1 and 2 let each message choose its own reliability.

MQTT through the years

1999

Invented by Andy Stanford-Clark (IBM) and Arlen Nipper to monitor oil pipelines over satellite links.

2010

IBM releases MQTT 3.1 royalty-free, opening it to every vendor.

2014

MQTT 3.1.1 becomes an OASIS standard. ISO/IEC 20922 follows in 2016.

2019

MQTT 5.0 adds reason codes, user properties and shared subscriptions.

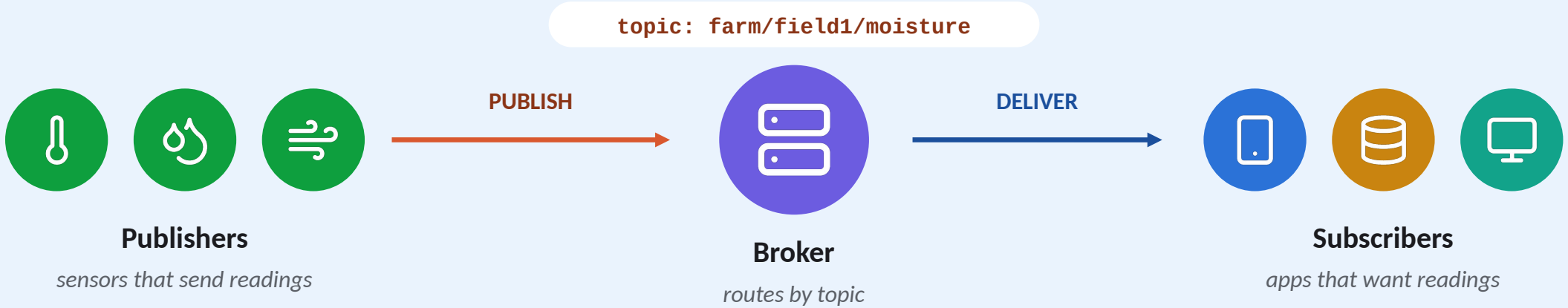
MQTT was built for unreliable, low-bandwidth links, which is exactly what IoT often gives us.

6

The publish-subscribe model

MQTT

Senders and receivers never talk directly. The broker sits between them.



Publisher

Sends a message to a topic. It does not know, or care, who is listening.

Broker

Receives every message and forwards it to the clients subscribed to that topic.

Subscriber

Tells the broker which topics it wants, then receives each matching message.

Why decouple?

Add a new subscriber without touching a single sensor. Nobody needs anyone's address.

Publishers and subscribers only need to know the broker and a topic name, never each other.

7

Topics and wildcards

MQTT

A topic is a path-like address. Wildcards let one subscription match many topics.

Anatomy of a topic



Levels are split by /. Topics are case-sensitive.

Nothing to set up: a topic exists as soon as someone publishes to it

Topics starting with $ (such as $SYS) are reserved for the broker

Keep names short and consistent: they travel with every message

+ Single-level wildcard

`farm/+/soil/moisture`

✓ `farm/field1/soil/moisture`	*matches*
✓ `farm/field2/soil/moisture`	*matches*
✗ `farm/field1/soil/temp`	*no match*

Multi-level wildcard

`farm/field1/#`

✓ `farm/field1/soil/moisture`	*matches*
✓ `farm/field1/valve/state`	*matches*
✗ `farm/field2/soil/moisture`	*no match*

Wildcards work only when subscribing. A publish always names one exact topic.

8

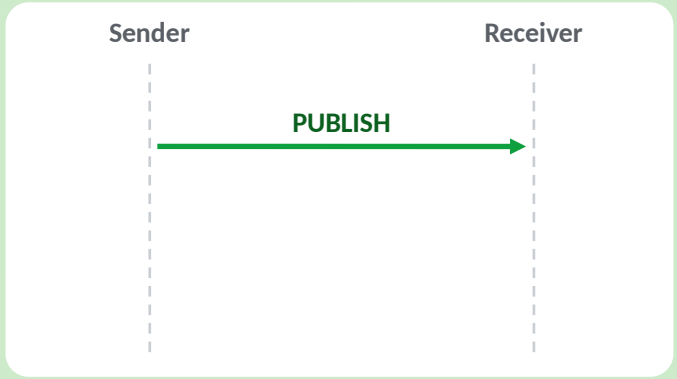
Quality of Service: three promises

MQTT

The sender picks how hard MQTT should try to deliver each message.

0 At most once

Fire and forget



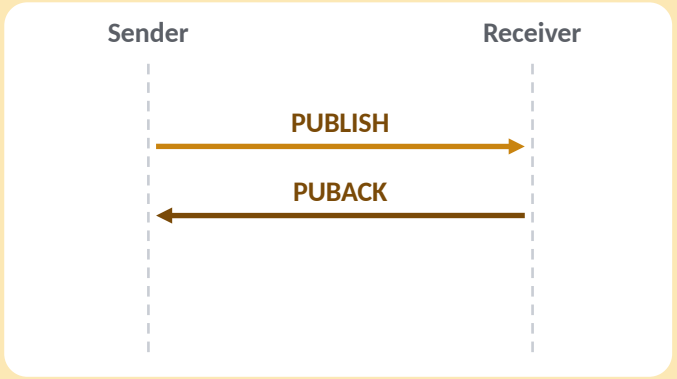
Packets per message 1

Best for

Frequent readings where losing one is fine, such as a temperature every second.

1 At least once

Acknowledged, may repeat



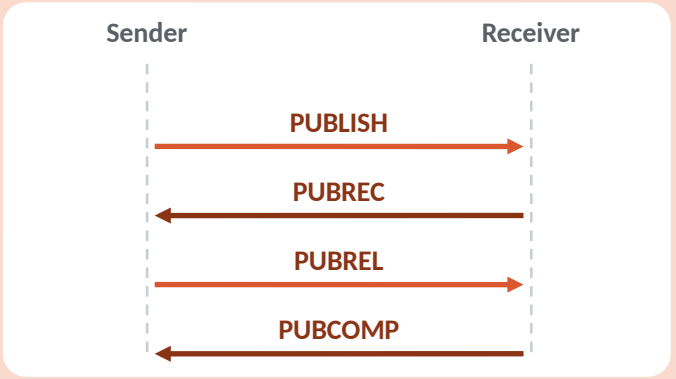
Packets per message 2

Best for

Readings and alerts that must arrive. A duplicate is possible, so handle repeats.

2 Exactly once

Four-step handshake



Packets per message 4

Best for

Commands that must never repeat, such as a billing event or an unlock trigger.

Higher QoS means more packets and more energy. Choose the lowest level the data can live with.

9

Built for flaky connections

MQTT

Four features that keep an MQTT system dependable when links drop.

Retained messages

The broker keeps the last message on a topic and hands it to every new subscriber at once.

```
farm/field1/valve/state = open    [retained]
```

Last Will and Testament

A message the client registers when it connects. If it drops off unexpectedly, the broker publishes it.

```
farm/field1/sensor/status = offline
```

Keep-alive

The client promises to send something within a set interval, using PINGREQ when idle. Silence means it is gone.

```
keep-alive = 60 s    PINGREQ / PINGRESP
```

Persistent sessions

The broker remembers subscriptions and queues QoS 1 and 2 messages while a client is offline.

```
Reconnect with the same client ID, then catch up
```

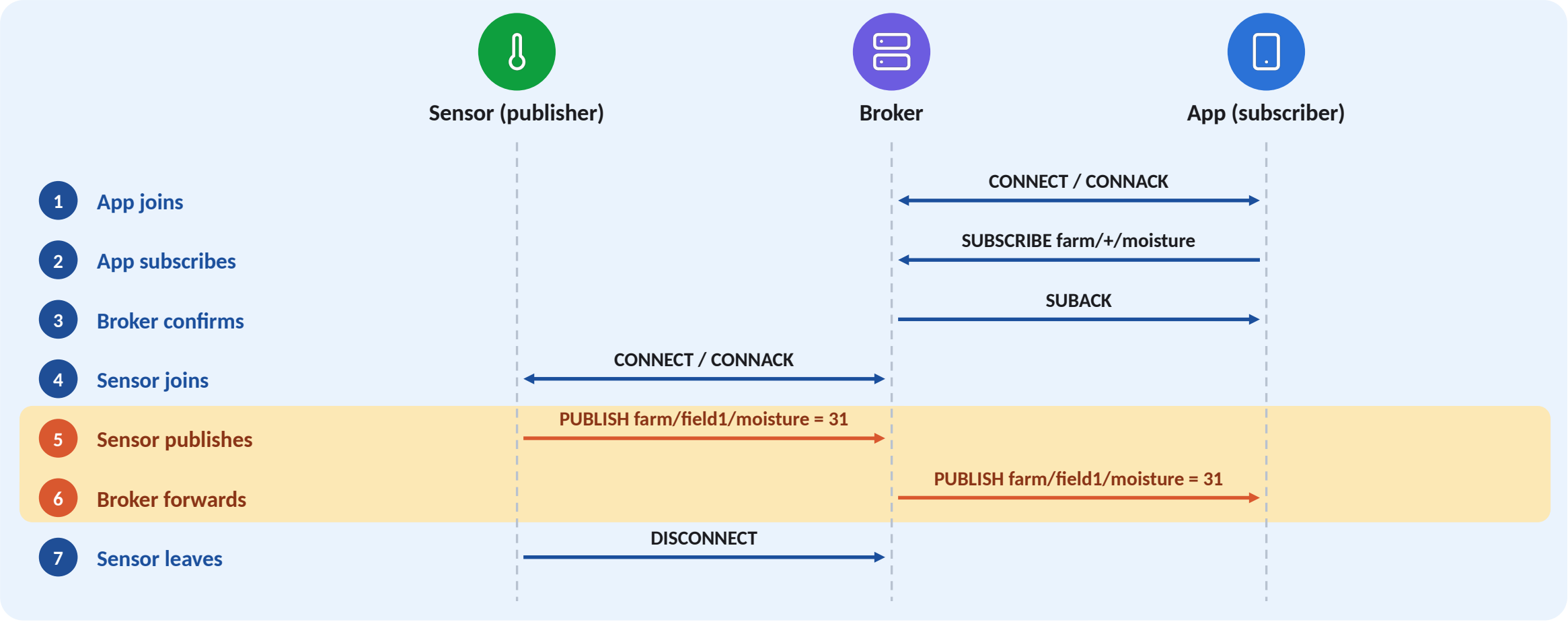
Together they give MQTT current status, failure alerts and catch-up, even on unreliable networks.

10

An MQTT session, step by step

MQTT

Every packet from joining to leaving, for one reading.



Only PUBLISH carries data. The rest is housekeeping that your MQTT client library does for you.

MQTT at a glance

Key facts to remember.

2 bytes
smallest fixed header

1883 / 8883
default TCP ports (8883 = TLS)

0, 1, 2
QoS levels per message

1 to many
one publish, many subscribers

Strengths

- Tiny header and small code: fits microcontrollers
- The broker decouples devices and scales to many clients
- QoS, retained messages and Last Will suit flaky links
- Wide support in open-source brokers and cloud IoT platforms

Watch-outs

- Needs a broker that you must run, secure and scale
- Keeps a TCP connection open: costly for deep-sleep devices
- Payload format is not defined: agree on JSON, CBOR or similar
- Plain MQTT is unencrypted: add TLS, authentication and access control

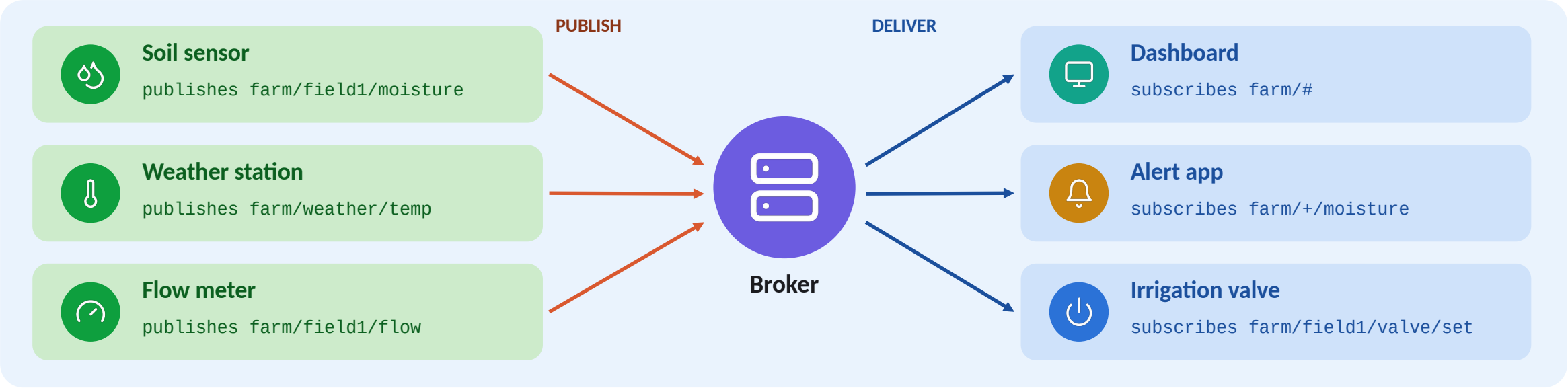
MQTT trades a broker and a persistent connection for simplicity, decoupling and scale.

11

MQTT in the field

MQTT

One broker can link a whole farm: sensors, dashboards and valves.



Where MQTT shines

Smart home
Lights, locks and thermostats

Smart agriculture
Soil, weather and irrigation

Industrial IoT
Machine telemetry and alarms

Connected vehicles
Location and status streams

Cloud platforms
Managed MQTT ingestion

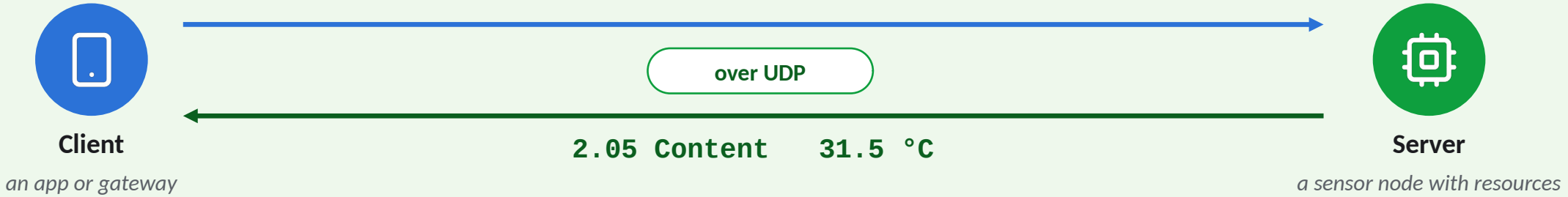
A dashboard user can also publish to farm/field1/valve/set, and the valve obeys within moments.

12

CoAP: REST for tiny devices

CoAP

Constrained Application Protocol: HTTP-style requests over UDP.



4 bytes

Fixed header

Small enough for the tiniest microcontrollers.

5683

Default UDP port

5684 when secured with DTLS.

4 verbs

GET, POST, PUT, DELETE

The same methods as HTTP, so REST ideas carry over.

RFC 7252

IETF standard, 2014

Designed for constrained nodes and networks.

CoAP keeps HTTP’s familiar request-response ideas but drops the weight.

13

Reliable on top of UDP

CoAP

CoAP adds its own delivery options, plus a way to watch a resource.

CON Confirmable

Please acknowledge. Resent with growing delays until an ACK arrives.

NON Non-confirmable

Fire and forget. No ACK, so the cheapest choice for frequent readings.

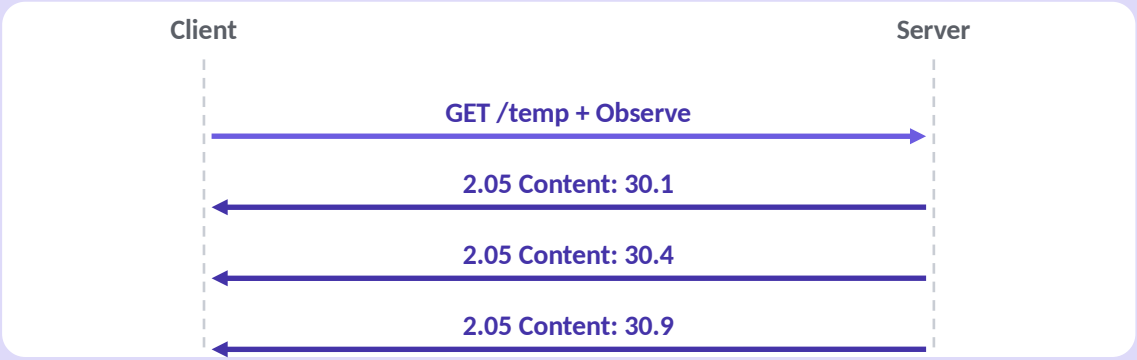
ACK Acknowledgement

Confirms a CON arrived. It can also carry the response itself.

RST Reset

Says a message could not be processed, or arrived without context.

Observe: watch a resource without a broker



Discovery

A GET on /.well-known/core lists what a device offers. Multicast finds devices on the network.

Block-wise

Large payloads, such as firmware, are sent as small numbered blocks.

CON gives reliability when you need it. NON keeps energy use minimal when you do not.

14

CoAP: strengths and limits

CoAP

Ideal for small devices that answer requests. Less suited to large fan-out.

Strengths

Tiny overhead
A 4-byte header suits very small microcontrollers

Low power
UDP has no connection to set up or keep alive

REST-friendly
Same verbs and URIs as HTTP; proxies can translate

Built-in extras
Observe, discovery and block-wise transfer

Limits

Unreliable base
UDP guarantees nothing; use CON and handle retries

Middlebox friction
NAT and firewalls often treat UDP less kindly

Smaller ecosystem
Fewer tools and cloud services than MQTT or HTTP

Point to point
No broker: fan-out relies on Observe or multicast

Typical uses

- Constrained sensors
- 6LoWPAN networks
- Smart lighting
- Device management (LwM2M)
- Building automation

Choose CoAP for small, sleepy devices that answer requests over UDP.

15

HTTP and HTTPS in IoT

HTTP / HTTPS

The web's own protocol: familiar, universal, and heavy for tiny devices.

DEVICE TO SERVER

```
POST /api/v1/readings HTTP/1.1
Host: iot.example.com
Content-Type: application/json
Authorization: Bearer <token>

{"device":"soil-07","moisture":31}
```

SERVER TO DEVICE

```
HTTP/1.1 201 Created
Location: /api/v1/readings/8841
```

Port 80: HTTP

Port 443: HTTPS (TLS)

REST + JSON

Why it is everywhere

Every language, browser, proxy and firewall understands it. REST and JSON make integration easy.

HTTPS = HTTP + TLS

Encrypts traffic and authenticates the server. Certificates and handshakes cost CPU, memory and battery on tiny devices.

One request, one response

The client always starts. A server cannot push data to a device without extra tricks.

HTTP is the easy default when a device is powerful, mains-powered and talks only occasionally.

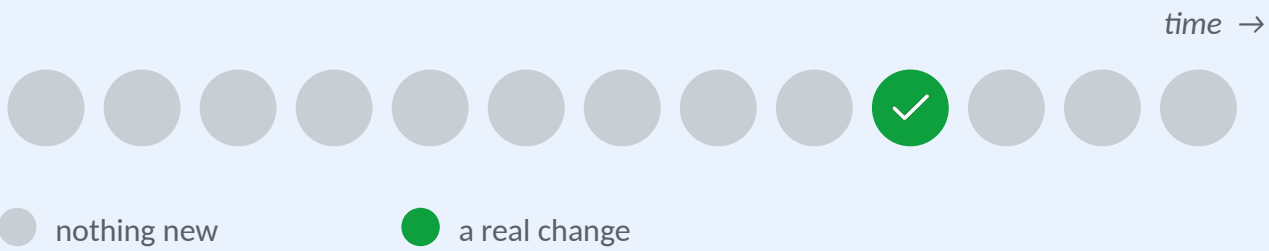
16

The polling problem

HTTP cannot push, so a device keeps asking: “anything new?”

HTTP / HTTPS

One poll every 10 seconds



8,640

requests a day at one poll every 10 seconds

86,400

requests a day at one poll every second

Each request carries full headers, and often a TLS session.

Ways around polling

Long polling

The server holds the request open until it has news.

Server-Sent Events

A one-way stream from server to client over HTTP.

WebSockets

Upgrade to a persistent two-way channel (coming up).

MQTT

The broker pushes messages to every subscriber.

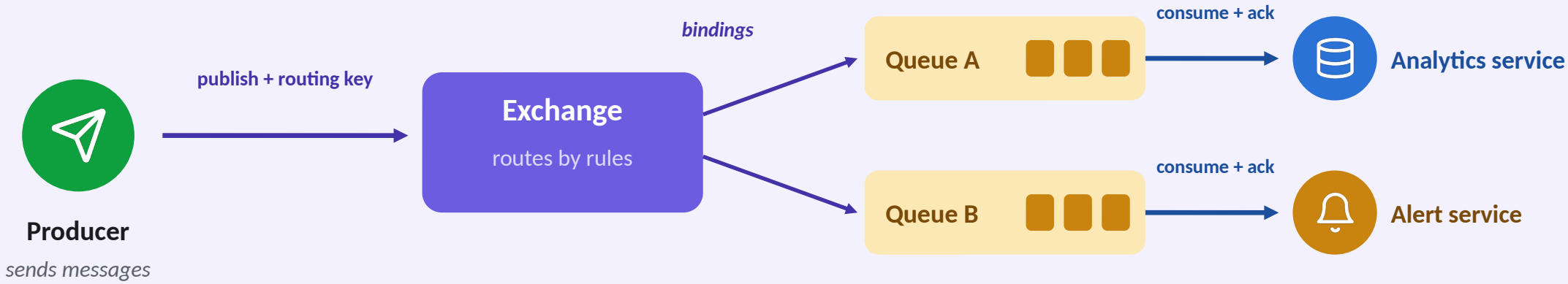
HTTP suits occasional calls. When a device must be reached, use a protocol built for push.

17

AMQP: queues with guarantees

AMQP

Advanced Message Queuing Protocol: routed, queued and acknowledged messaging.



Exchange types

Direct, fanout, topic and headers: four ways to route a message.

Durable queues

Messages wait in a queue until a consumer acknowledges them.

Binary over TCP

Port 5672, or 5671 with TLS. Compact frames with an 8-byte header.

Two versions

0-9-1 is used by RabbitMQ. 1.0 is an OASIS and ISO standard, used by Azure services.

AMQP does the routing and the remembering, so producers and consumers can be slow, offline or busy.

18

AMQP: reliable, but heavier

AMQP

Built for back-end systems that cannot afford to lose a message.

What AMQP guarantees

Acknowledged delivery

A message stays queued until the consumer confirms it.

Durable storage

Persistent queues and messages survive a broker restart.

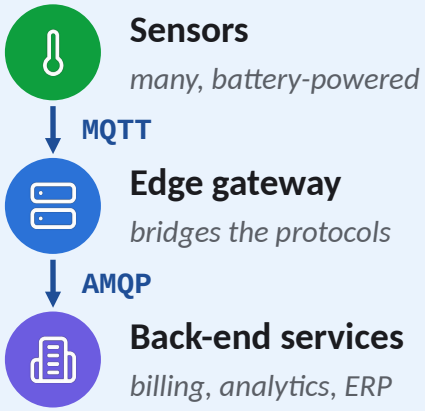
Flexible routing

One message can reach many queues, chosen by rules.

Flow control

Consumers receive messages at a pace they can handle.

A common pairing



Watch-outs

- Heavier than MQTT: larger frames, more state
- Broker setup and tuning take effort
- Overkill for a single small sensor

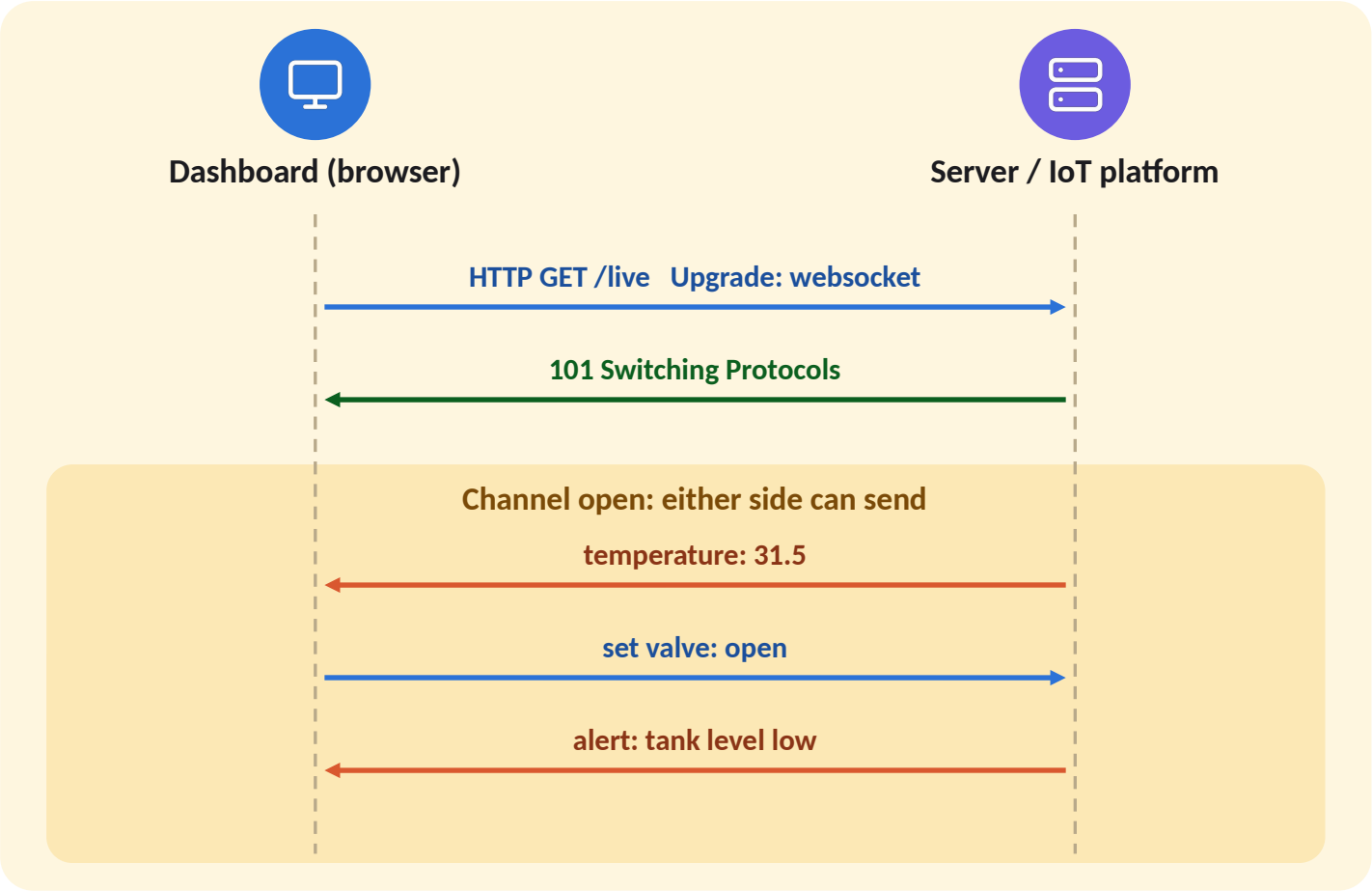
MQTT at the edge, AMQP in the back-end: a common pairing in real IoT systems.

19

WebSockets: one connection, two ways

WebSockets

RFC 6455: starts as HTTP, then upgrades to a persistent two-way channel.



ws:// and wss://

Default ports 80 and 443. The wss form is secured with TLS, like HTTPS.

Tiny frames

After the handshake, each message adds only 2 to 14 bytes of framing.

Either side can speak

No polling: the server can push a reading the moment it arrives.

Set up once, then talk freely both ways: no polling and no repeated headers.

20

WebSockets: where they fit

WebSockets

Excellent for live screens. Not a replacement for a message broker.

Strengths

- Real-time push in both directions
- Works in every browser
- Starts as HTTP, so it usually passes firewalls
- One connection replaces repeated polling

Watch-outs

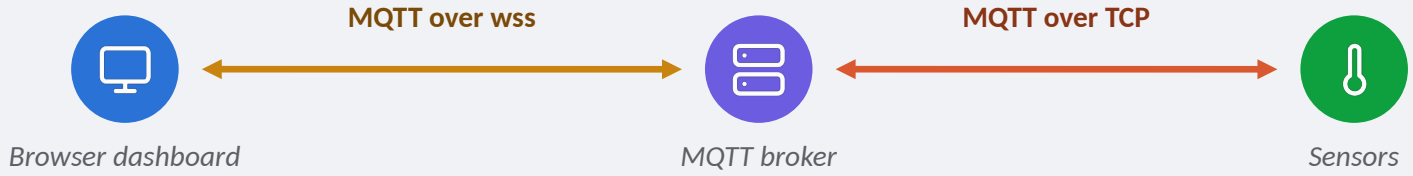
- No built-in topics or QoS: you design the message rules
- The connection must stay open: hard on sleeping devices
- Needs reconnect logic when links drop
- Many open connections use server resources

Where it fits

- Live dashboards and control panels
- Browser access to MQTT brokers
- Real-time alerts and notifications
- Gateways streaming telemetry to cloud services

MQTT over WebSockets

Browsers cannot open raw TCP sockets, but they can open WebSockets. So a web dashboard can subscribe to a broker over wss.



Use WebSockets when a screen needs to react the moment data changes.

Side by side

Five protocols, six questions. Read down a column to meet one protocol.

Feature	MQTT	CoAP	HTTP/HTTPS	AMQP	WebSockets
Transport	TCP	UDP	TCP (TLS for HTTPS)	TCP	TCP, after an HTTP upgrade
Message overhead	2-byte fixed header	4-byte fixed header	200+ bytes of text headers	8-byte frame header	2 to 14 bytes per frame
Pattern	Publish / subscribe through a broker	Request / response, plus Observe	Request / response	Queues, exchanges and routing	Persistent two-way channel
Reliability	QoS 0, 1 or 2	CON with retries, or NON	Whatever TCP gives	Acknowledged, durable queues	None built in
Default port	1883 / 8883 (TLS)	5683 / 5684 (DTLS)	80 / 443	5672 / 5671 (TLS)	80 / 443 (ws / wss)
Best use-case	**Sensor telemetry at scale**	**Tiny, sleepy devices**	**Occasional calls from strong devices**	**Reliable back-end pipelines**	**Live dashboards and control**

There is no single winner. Each protocol is built for a different shape of conversation.

21

How to choose

Decision guide

A quick decision guide, in five questions.

1. **Does a web page need to update the moment data changes?**
   - Yes → **WebSockets** — A live two-way channel to the browser.
   - No ↓
2. **Must back-end services never lose a message, with routing rules?**
   - Yes → **AMQP** — Durable queues and acknowledged delivery.
   - No ↓
3. **Is the device tiny and sleepy, answering simple requests on a light network?**
   - Yes → **CoAP** — HTTP-style requests in a 4-byte header.
   - No ↓
4. **Is the device powerful, and does it talk only now and then?**
   - Yes → **HTTP / HTTPS** — Universal, easy to build and to secure.
   - No ↓
5. **Many devices, small readings, patchy links?**
   - Yes → **MQTT** — The IoT default: light, brokered, with QoS.

A teaching shortcut: real designs also weigh device power, security, cost and the tools you already have.

Which one would you pick?

Decide first, then read the answers.

1 **A greenhouse has 200 battery sensors on patchy Wi-Fi, sending a small reading every minute to a dashboard.**

MQTT

Tiny headers, a broker to fan out, QoS 1 for patchy links.

2 **A lamp in a low-power mesh network must switch on when a controller sends a simple request.**

CoAP

REST-style GET/PUT over UDP suits constrained nodes.

3 **A mains-powered gateway uploads one summary a day to a cloud REST API.**

HTTPS

Rare, powerful, and needs universal support: HTTP wins.

4 **A factory sends order events to billing. None may be lost, even if the billing service is down.**

AMQP

Durable queues keep messages until they are acknowledged.

5 **An operator's browser must show a tank level the instant it changes and open a valve.**

WebSockets

A persistent two-way channel with no polling.

6 **Field sensors report to the edge, and the edge feeds reliable back-end services.**

MQTT + AMQP

Light at the edge, guaranteed behind it: a common pairing.

Ask: who talks to whom, how often, how reliably, and on what kind of device?

The big picture

Five protocols in one line each.

MQTT	Publish and subscribe through a broker. Two-byte header, three QoS levels: the IoT default.
CoAP	REST for tiny devices over UDP, with optional confirmation and Observe.
HTTP / HTTPS	The web's protocol: universal and easy, but heavy, and the client must ask.
AMQP	Exchanges, queues and acknowledgements for back-end systems that cannot lose data.
WebSockets	One upgraded connection, two-way and instant: ideal for live screens.
Up next	Topic 5: IoT networking concepts: IPv4 and IPv6, 6LoWPAN, WSN and gateways.

Radios move the bits. Protocols give them meaning. Match the protocol to the message.

One-line takeaway

“The best protocol is the one that fits the conversation. Use MQTT when many devices report, CoAP when they are tiny, and HTTP, AMQP or WebSockets when the job demands them.”

MQTT | CoAP | HTTP/HTTPS | AMQP | WebSockets

Up next: Topic 5, IoT networking concepts: IPv4 and IPv6, 6LoWPAN, wireless sensor networks and gateways.

1

IoT networking: the reference picture

Five concepts that turn an isolated sensor into a reachable network citizen

In Simple Words

IoT networking concepts are the rules that let a constrained device get an address, join a network, and have its data routed to wherever it is needed — even with almost no power, memory, or radio range to spare.

Five concepts run through this whole topic:

1. IPv4 / IPv6 addressing
2. 6LoWPAN compression
3. Wireless Sensor Networks
4. Gateways
5. Addressing & routing (RPL)

Why We Need These Concepts

Scale

Tens of billions of devices each need their own reachable identity.

Constrained devices

Tiny batteries, memory, and radios rule out heavyweight protocols.

Multi-hop reach

Most sensors can't reach the internet directly — they relay through neighbours.

Interoperability

Zigbee, BLE, LoRa and Wi-Fi devices all need a common way to connect.

Learn these five once, and you can read the network diagram of almost any IoT deployment.

2 Why does addressing matter so much today?

The device count is climbing faster than the old address space can handle

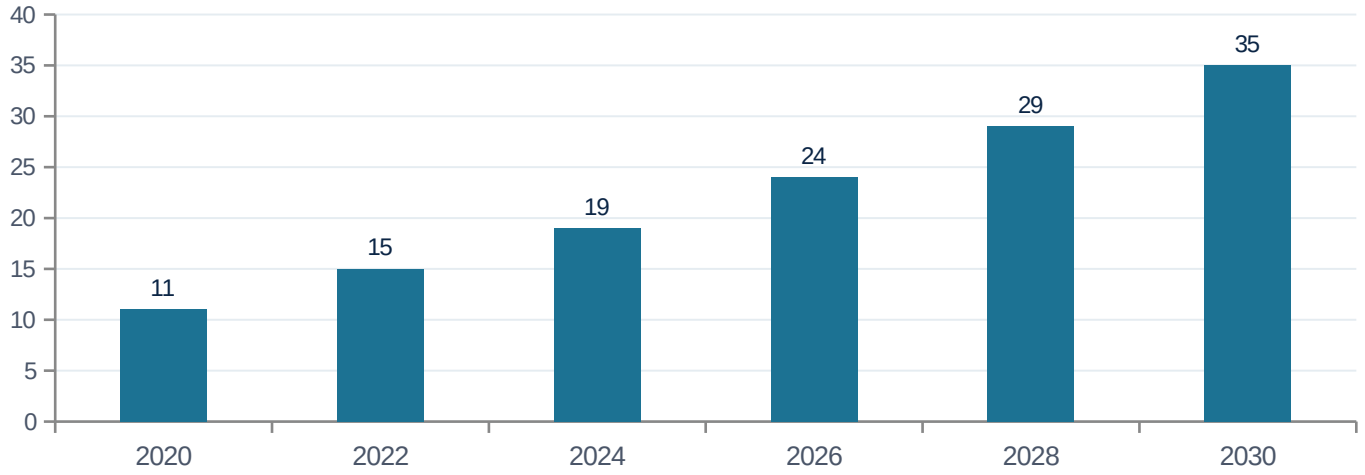
Running out of room

IPv4 offers about 4.3 billion addresses. That pool was exhausted years ago — long before IoT began adding billions of new devices.

Every “thing” needs an identity

A device with no address can’t be reached remotely. No address means no monitoring, no control, and no place on the network.

Illustrative growth in connected IoT devices (worldwide, billions)



Illustrative, not a forecast

Exact figures vary by source and year, but every credible estimate agrees on the shape of the curve: device counts trend sharply upward, and IPv4’s 4.3 billion addresses were never going to keep up.

Concept 1 · IPv4 and IPv6 in IoT

The address space IoT is built on — and why it had to grow

IPv4 — the legacy net

Address size

32-bit → about 4.3 billion addresses

Header

Fixed fields, checksum recalculated at every hop

Status in IoT

Exhausted — devices sit behind NAT, adding translation overhead

Practical effect

Hard to give every sensor its own public address

IPv6 — built for scale

Address size

128-bit → about 340 undecillion addresses

Header

Simplified, fixed-size 40-byte header, extension-based options

Status in IoT

Every device can get a unique, global address — no NAT needed

Practical effect

Native support for 6LoWPAN compression on tiny radios

IoT's sheer scale of billions of constrained devices is exactly why IPv6 adoption accelerated.

Concept 2 · 6LoWPAN squeezes IPv6 onto tiny radios

IPv6 over Low-power Wireless Personal Area Networks



IPv6 packet

Large header, up to 1280-byte MTU



Header compression

Strips redundant IPv6/UDP header fields



Fragmentation

Splits packets to fit tiny frames



802.15.4 frame

Only 127 bytes on the air

Why not just use IPv6 as-is?

- An IPv6 header alone can be 40 bytes — nearly a third of an 802.15.4 frame’s 127-byte budget
- Constrained radios can’t afford that overhead on every single packet
- 6LoWPAN was created specifically to close this gap without abandoning IPv6
- Result: tiny nodes keep a real, globally routable IP address



In practice

A coin-cell battery sensor can now carry a full, globally routable IPv6 address instead of a proprietary short code — and talk straight to the internet.

3

How 6LoWPAN actually does its job

Four jobs bolted onto the IPv6 stack for constrained radios



Header compression

IPHC (IPv6 Header Compression) elides fields that can be inferred from context — link-layer addresses, hop limits, well-known prefixes — instead of sending them in full.



Fragmentation & reassembly

Packets that still exceed 127 bytes are split into a chain of fragments, each tagged with a datagram tag and offset, and reassembled at the receiving node.



Mesh addressing

Nodes route through each other using link-layer mesh headers, so a reading can hop several nodes before reaching a point with internet access.



Low-power operation

Nodes sleep between transmissions and wake on a schedule, so 6LoWPAN's framing has to stay cheap to encode and decode on a tiny microcontroller.

Concept 3 · Wireless Sensor Networks (WSN)

Hundreds to thousands of battery-powered nodes sensing at scale



Sensing unit

Measures temperature, moisture, motion…



Processing unit

Local logic on a tiny microcontroller



Transceiver

Short-range radio to neighbouring nodes



Power unit

Small battery, often solar-assisted



Sink / base station

Collects data and forwards it to a gateway

Star topology

Every node talks directly to one central hub. Simple, but the hub’s range limits how far the network reaches.

Mesh topology

Nodes relay through each other. Reaches much farther and survives a dead node, at the cost of routing complexity.

4

What makes wireless sensor networks hard

The same three constraints show up in almost every real deployment



Unreliable links

Wireless links drop packets, suffer interference, and vary with weather and obstacles — a reading may need several attempts to get through.



Limited power

Nodes often run for years on a single battery or small solar cell, so every extra transmission or computation has a real cost.



Huge distances

A deployment can span a farm, a city, or a national grid — far beyond what any single radio hop can cover.



Why it matters: these three constraints — unreliable links, limited power, huge distances — are exactly why WSNs need their own addressing, routing, and mesh strategies instead of borrowing wired-network designs outright.

Concept 4 · Gateways bridge two worlds

Every IoT gateway sits between a low-power local network and the internet

Short-range side

Zigbee, BLE, 6LoWPAN and LoRa radios talk to nearby sensors and actuators using tiny, low-power frames.

translates protocols · aggregates data · often filters or pre-processes before sending on

Internet side

Wi-Fi, Ethernet, or cellular (4G/5G, NB-IoT) then carry the aggregated data onward to the cloud.

5 Where should processing intelligence live?

A design question that runs through every one of these five concepts

On the node

Closest to the sensor

WHERE

Right on the constrained device itself, in the same microcontroller that reads the sensor.

WHAT IT DOES

Runs the lightest possible check — a threshold, a simple filter — before anything is even sent.

IMPACT

Lowest latency and radio use, but very little memory or compute to work with.

At the gateway

One hop out

WHERE

On the gateway that already bridges the local network to the internet.

WHAT IT DOES

Aggregates readings from many nodes, deduplicates, and filters before forwarding.

IMPACT

A local “middle manager” that spares the cloud a raw firehose of tiny packets.

In the cloud

Farthest from the device

WHERE

A distant, centralized data center reached over the wider internet.

WHAT IT DOES

Runs the heaviest analysis — cross-device correlation, long-term trends, model training.

IMPACT

Most powerful, but adds latency and depends on every hop staying connected.

Concept 5 · Addressing and routing

Every node needs a name — and a path that gets its data where it needs to go

Addressing

Every node in the network gets a unique identifier — typically a compressed IPv6/6LoWPAN address — so it can be located individually, even inside a mesh of thousands of neighbours.

Routing

RPL (Routing Protocol for Low-power and Lossy Networks) builds a tree — a DODAG — rooted at the gateway, so every node knows the lowest-cost next hop toward it.

A DODAG rooted at the gateway



6 How one reading actually hops to the gateway

Following a single packet through the RPL tree, step by step



1. Sense

A leaf node takes a reading and prepares a 6LoWPAN packet.



2. Look up route

It checks its routing table for the best next hop toward the root.



3. Hop, hop, hop

Each intermediate node repeats the same lookup and forwards it onward.



4. Reach the gateway

The DODAG root receives the packet and passes it onto the internet side.



5. Acknowledge

An acknowledgment can flow back down the same path, confirming delivery.

Example: a leaf sensor's reading hops through three neighbouring nodes, each one forwarding it along the RPL tree, before it reaches the gateway.

7 Worked example: mapping a real deployment

A smart-agriculture sensor network, placed onto all five concepts

Concept	What it looks like on the farm
IPv4 / IPv6	Each soil sensor is assigned a compressed IPv6 address so it can be reached individually
6LoWPAN	That IPv6 address and each reading are compressed to fit the 802.15.4 radio frame
WSN	Dozens of soil-moisture nodes form a mesh, relaying readings toward a sink at the field's edge
Gateway	A field gateway aggregates the mesh's data and forwards it over cellular to the cloud
Addressing & routing	RPL keeps every node's route to the gateway up to date as conditions and battery levels change

Once this clicks, you can run the same five-concept mental exercise on any IoT deployment you meet.

8 IoT networking concepts at a glance

Concept · role · a concrete example

Concept	Role	Example
IPv4	Legacy 32-bit addressing, now exhausted	*Still carries most of today's internet*
IPv6	Massive address space for billions of devices	*Every streetlight gets its own address*
6LoWPAN	Compresses IPv6 to fit low-power radios	*A battery sensor carries a full IP address*
WSN	Many sensor nodes cooperating wirelessly	*Soil-moisture network across a farm*
Gateway	Bridges the local network to the internet	*Smart-home hub, cellular IoT gateway*
Routing (RPL)	Finds efficient multi-hop paths	*Readings hop node-to-node to a sink*

The big picture

Five concepts, one job: get every device addressed, connected, and reachable

5 concepts	IPv4/IPv6, 6LoWPAN, WSN, Gateways, Addressing & Routing — the full pipeline.
Every device	Even a coin-cell sensor can carry a real, globally routable address.
Node · Gateway · Cloud	Where processing happens is a recurring design decision.
Any deployment	The five-concept mental exercise maps onto any IoT system you meet.
Up next	Unit II wraps up with a full side-by-side comparison and recap.



One-line takeaway

“IoT networking concepts — addressing, compression, meshing, and gateways — turn billions of tiny, power-starved devices into full citizens of the internet.”

IPv4 & IPv6 | 6LoWPAN | WSN | Gateways | Addressing & Routing

Up next: Unit II Comparison Table & Recap

1

Five topics, one system

Everything data needs to travel from a sensor to the cloud, and back

In Simple Words

Unit II covered how data actually gets from a device to the people and systems that use it: the short links to nearby devices, the long links across a city or country, the protocols that shape each conversation, and the addressing and routing plumbing that ties it all together.

Twenty-plus technologies. Four questions asked of each: range, power, bandwidth, use-case.

The Five Topics

1 · Introduction

Why communication is IoT's backbone

2 · Short-range

Bluetooth/BLE, Zigbee, RFID, NFC, Wi-Fi

3 · Long-range

LoRaWAN, NB-IoT, LTE-M, 4G/5G

4 · Protocols

MQTT, CoAP, HTTP/HTTPS, AMQP, WebSockets

5 · Networking concepts

IPv4/IPv6, 6LoWPAN, WSN, gateways, routing

Short-range technologies, side by side

Topic 2 recap — typical values only, always check the datasheet

Technology	Band	Typical range	Data rate	Power	Topology
Bluetooth LE	2.4 GHz	Tens of meters	Up to 2 Mbps	Very low	Point-to-point, broadcast, mesh
Zigbee	2.4 GHz (also sub-GHz)	Tens of m per hop	Up to 250 kbps	Low	Mesh
RFID	LF, HF or UHF	cm to several meters	Low (ID reads)	Passive tags: none	Reader to tag
NFC	13.56 MHz	A few centimeters	Up to 424 kbps	Very low	Tap: peer-to-peer or tag
Wi-Fi	2.4 / 5 / 6 GHz	Tens of m indoors	Mbps to Gbps	Higher	Star via access point

No single technology wins everywhere: match the link to the application.

Long-range technologies, side by side

Topic 3 recap — range, power, bandwidth and use-case

	LoRaWAN	NB-IoT	LTE-M	4G/5G
Spectrum	Unlicensed sub-GHz (865–867 MHz in India)	Licensed cellular	Licensed cellular	Licensed cellular
Range	2–5 km urban, 15 km+ rural	Cell coverage, deep indoors	Cell coverage	Cell coverage
Power	Up to ~10 years on a battery	10+ years (design target)	Years on a battery	Highest draw; often mains-powered
Bandwidth	≈ 0.25–50 kbps	≈ 20–250 kbps	Up to ~1 Mbps	10 Mbps to Gbps
Mobility	Limited	Fixed	Yes, with handover	Yes
Best use-case	Farms, campuses, metering	Fixed meters, parking	Tracking, wearables	Video, vehicles, factories

No single winner: LoRaWAN buys freedom, NB-IoT and LTE-M buy carrier reach, 4G/5G buys bandwidth.

Protocols, side by side

Topic 4 recap — five protocols, six questions

	MQTT	CoAP	HTTP/S	AMQP	WebSockets
Transport	TCP	UDP	TCP (TLS for HTTPS)	TCP	TCP, after an HTTP upgrade
Message overhead	2-byte fixed header	4-byte fixed header	200+ bytes of text headers	8-byte frame header	2–14 bytes per frame
Pattern	Publish/subscribe via a broker	Request/response, plus Observe	Request/ response	Queues, exchanges, routing	Persistent two-way channel
Reliability	QoS 0, 1 or 2	CON with retries, or NON	Whatever TCP gives	Acknowledged, durable queues	None built in
Default port	1883 / 8883 (TLS)	5683 / 5684 (DTLS)	80 / 443	5672 / 5671 (TLS)	80 / 443 (ws / wss)
Best use-case	Sensor telemetry at scale	Tiny, sleepy devices	Occasional calls, strong devices	Reliable back-end pipelines	Live dashboards and control

No single winner: each protocol is built for a different shape of conversation.

Networking concepts, side by side

Topic 5 recap — concept, role, and a concrete example

Concept	Role	Example
IPv4	Legacy 32-bit addressing, now exhausted	Still carries most of today's internet
IPv6	Massive address space for billions of devices	Every streetlight gets its own address
6LoWPAN	Compresses IPv6 to fit low-power radios	A battery sensor carries a full IP address
WSN	Many sensor nodes cooperating wirelessly	Soil-moisture network across a farm
Gateway	Bridges the local network to the internet	Smart-home hub, cellular IoT gateway
Routing (RPL)	Finds efficient multi-hop paths	Readings hop node-to-node to a sink

Addressing, compression, meshing, and gateways — the plumbing beneath every link and protocol above.

2 How to choose: a unified decision guide

Four questions, asked in order, cover the whole unit



1 · How far?

Same room, same building, or across a city and beyond?

Short-range (Topic 2) if it's nearby. Long-range (Topic 3) if it isn't.



2 · What shape is the message?

One-off request, a live stream, or a broker fan-out to many subscribers?

That shape points to a protocol (Topic 4): MQTT, CoAP, HTTP, AMQP, or WebSockets.



3 · How much power and money?

Coin-cell for years, or mains-powered and unconstrained?

Power budget narrows the field fastest, in both the link and the protocol layer.



4 · How many devices, how addressed?

A handful of gateways, or thousands of individually addressed nodes?

Scale is exactly what Topic 5's networking concepts — IPv6, 6LoWPAN, routing — are built for.

3 Quick self-assessment

Five questions to check what stuck — answers use everything from Topics 1 to 5

1. A battery sensor needs to run for years and only sends a few bytes a day from a remote farm. Which long-range option first?

2. A thousand sleepy devices publish tiny readings to one broker. Which protocol, and why not HTTP?

3. A smartwatch needs to pair with a phone for a few hours of battery life. Which short-range link?

4. Why can’t a coin-cell sensor just use IPv6 headers unmodified over 802.15.4?

5. A gateway aggregates fifty mesh nodes. What two networking concepts is it responsible for?

The big picture

Unit II in five rows: everything that gets a device's data where it needs to go

Short-range	Bluetooth/BLE, Zigbee, RFID, NFC, Wi-Fi — the last few meters to a gateway.
Long-range	LoRaWAN, NB-IoT, LTE-M, 4G/5G — kilometers, traded against power and speed.
Protocols	MQTT, CoAP, HTTP/HTTPS, AMQP, WebSockets — the shape of the conversation.
Networking	IPv4/IPv6, 6LoWPAN, WSN, gateways, routing — addressing at any scale.
Up next	Unit III: IoT Platforms, Edge and Cloud — where this data actually gets used.



One-line takeaway

“There is no single best technology — only the one that fits how far, how fast, how often, and how many devices you’re talking to.”

Short-range | Long-range | Protocols | Networking | Self-check

Up next: Unit III — IoT Platforms, Edge and Cloud (Arduino, ESP32, Raspberry Pi, cloud & edge computing, IoT + AI)